

MAPPING FOREST GOVERNANCE: GEOSPATIAL SUPPLY CHAIN
ANALYSIS OF FOREST CODE IMPLEMENTATION IN MATO GROSSO

by

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Abstract

This thesis presents a geospatial and statistical evaluation of the Brazilian Forest Code's (FC) implementation in the state of Mato Grosso, with a specific focus on the cattle production chain. Utilizing geoprocessing tools to analyze 176,020 registered rural properties covering 82.3 million hectares, the study quantifies environmental compliance with the law regarding Legal Reserves (LR) and Permanent Preservation Areas (APP), applying the rules that defined lower LR requirements up to the year 2000 and benefits for consolidated areas up to 2008. The results indicate that while the state possesses an overall LR vegetation surplus of 13.8 million hectares, 43,571 properties still carry a net LR deficit of 4.51 million hectares. Of this deficit, 3.18 million hectares can be regularized through off-site compensation, while 1.33 million hectares require mandatory on-site restoration. Additionally, 1.13 million hectares of APPs require on-site restoration. By cross-referencing environmental data with Animal Transit Guides (GTA), we show that the most active direct suppliers account for less than 15% of the total environmental liability, while the majority of the restoration obligations (65.7%) are found on properties that participate less frequently or less actively in the final segments of the supply chain. The concentration of the Mato Grosso cattle sector's deficits in potentially more marginal portions of the supplier base portends a potential challenge to motivating major impacts on state-level deficits through cattle supply chain policies. The study concludes that achieving full compliance with the Forest Code requires the acceleration of the CAR validation, individual cattle traceability, and the implementation of tools like the Green Passport Program and environmental reserve quotas.

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1. Introduction

This master's thesis contributes to the fundamental debate on sustainable land use in Brazil—a country defined by the paradox of being a megadiverse nation, home to the world's largest expanse of tropical forests (Mongabay, 2024), while also functioning as a global agricultural powerhouse. Since 2023, Brazil has been the world's largest exporter of agricultural commodities, with beef and soybeans as its leading products, together accounting for approximately 54% of Brazilian agribusiness exports (Cardoso, 2025).

However, the expansion of the agricultural frontier to enable commodity production at this scale has been historically associated with high rates of deforestation. Between 1985 and 2024, the country lost approximately 13% of its natural cover (Mapbiomas, 2025), with cattle ranching being the primary driver for the conversion of forests into pastures, accounting for over 80% of Brazil's deforestation (Skidmore et al., 2021; Rajão et al., 2020). In a context of global climate urgency, the persistent loss of native vegetation represents a direct threat to the fulfillment of national Greenhouse Gas (GHG) mitigation targets, given that land-use changes were responsible for 42% of Brazil's emissions in 2024 (SEEG, 2025). Deforestation also poses a challenge to biodiversity conservation across all biomes (Brock et al., 2021; Soares-Filho et al., 2014). Concurrently, total global demand for food is projected to increase around 35% to 56% between 2010 and 2050 (van Dijk et al., 2021). Reconciling the demand for food with the preservation of natural resources has become a crucial, pressing issue.

Brazil is governed by some of the strongest and most innovative forest conservation policies in the world, and its legal regime with regard to environmental protections even on private lands is unique. The 1988 Brazilian Federal Constitution establishes that the right to property is not absolute and must respect the environment (Brazil, 1988). This builds on a longstanding legislative tradition of prioritizing the protection of natural resources dating back to the first Forest Code of 1934, which mandated the

preservation of native vegetation on private land as a legal obligation and the responsibility of landowners (Brazil, 1934). As a result, a significant amount of Brazil's protected areas are under the control of private landowners.

This is markedly different than the legal regime regarding conservation of the United States of America (USA) for example, where conservation on private properties is largely voluntary and typically incentivized by financial compensation (Chivari et al., 2017). In fact, comparative research examining forest legislation in countries including the United States, China, Canada, France, Germany, and Argentina found that Brazil is the only country, among those analyzed, that requires the maintenance of a fixed percentage of native vegetation on all private properties, and mandates riparian protection zones are significantly larger than those required in most other countries (Chivari et al., 2017). What's more, protection of these private reserves is taken very seriously in Brazil; within the Brazilian legal system, unauthorized removal of native vegetation subjects the offender to a triple liability, meaning that offenders may face simultaneous administrative, civil, and criminal sanctions (Brazil, 2012). As of 2024, approximately 67% of Brazil's territory remained under forest cover (Mapbiomas, 2025), a level of preservation that is closely linked to this robust legal and regulatory framework.

The Forest Code (FC), Law No. 12,651/2012, is Brazil's main legal instrument for reconciling agricultural production on private properties with nature conservation. The FC requires the maintenance of two key conservation areas on all rural properties: Permanent Preservation Areas (Áreas de Preservação Permanente, APPs) and the Legal Reserve (Reservas Legais, RLs) (Chiavari et al., 2023). The RL requirement varies by biome and property location, reaching 80% in forest areas in the Amazon, 35% in the Cerrado in states that are part of the Legal Amazon, and 20% in other Brazilian biomes (Stickler et al., 2013). The FC also requires rural landowners to obtain prior authorization from state environmental agencies before clearing native vegetation, ensuring that deforestation does not occur within APPs or RLs (Brazil, 2012). Failure to follow this procedure renders deforestation illegal, which can result in multiple sanctions, including administrative embargo on land use, mandatory restoration of the deforested area, financial penalties, and prosecution under environmental criminal law (Brazil, 1998; Brazil, 2012). If the property has vegetation exceeding the Permanent Preservation Area (APP) and Legal Reserve (RL) requirements, the surplus may constitute an Environmental Reserve Quota (CRA).

This quota can be traded with farmers who have an RL deficit in consolidated areas, defined by law as those deforested prior to July 22, 2008.

The operationalization and enforcement of the FC relies on the Rural Environmental Registry (Cadastro Ambiental Rural, CAR), a mandatory electronic public registry that functions as the central database for monitoring, control, and environmental planning of land use (Luiz, 2022). However, the full implementation of the FC faces significant barriers, notably the slow pace of the analysis and validation stage for the self-declared data submitted to the CAR. More than a decade after the law's enactment, only 6% of the national registries were validated by the end of 2024 (Lopes, 2025).

Given the complexity of the Forest Code rules, it is only upon the validation of CAR data that the landowner obtains legal certainty regarding which areas may be used for agricultural production and which must be preserved. The CAR also serves as the entry point for the Environmental Regularization Program (Programa de Regularização Ambiental, PRA), which enables landowners with identified environmental liabilities to commit to restoring of APPs and RLs (Chiavari et al., 2023).

Although Brazil possesses one of the most stringent private land conservation frameworks in the world, translating these legal mandates into on-the-ground compliance is hindered by systemic administrative bottlenecks. Recent data indicates an improvement in the government's capacity to detect and initially respond to deforestation; in 2024, approximately 54% of the deforested area in Brazil intersected with some form of state action, whether through prior authorization or enforcement measures such as embargoes (MapBiomass, 2025). In Mato Grosso, central focus of this work, the state environmental agency (SEMA-MT) achieved a higher response rate, taking action on 59.1% of the deforested area (MapBiomass, 2025).

However, issuing an infraction is merely the first step, and the subsequent administrative and judicial phases expose a severe enforcement gap. Historically, the traditional command-and-control approach has suffered from high rates of impunity. At the federal level, only one in ten fines issued is paid (Bechara, 2024). This bottleneck persists acutely at the subnational level. An analysis of SEMA-MT's administrative proceedings revealed that between 2017 and 2021, the agency lost R\$ 117.6 million to expired environmental fines due to extreme procedural delays (Bolzani, 2022). This highlights the need to integrate market-driven mechanisms with economic consequences for non-compliant agents,

reshaping the governance landscape.

The role of socio-environmental controls through agreements with stakeholders.

Market-based agreements in Brazil's rural sector have historically signaled that illegal deforestation should result in restrictions on commercial opportunities. Initiatives such as the Conduct Adjustment Agreement (MPF-TAC) and the Public Livestock Commitment (PLC, formerly known as G4), implemented starting in 2009, require slaughterhouses operating in the Amazon biome to block the purchase of cattle from properties with illegal deforestation or environmental embargos, or lacking a CAR registration (Gibbs, 2015).

Evidence suggests that these agreements have had some measurable impacts. In the state of Pará for example, the probability of major slaughterhouses like JBS purchasing from properties with recent deforestation fell considerably following their implementation (Gibbs, 2016). Similarly, the Soy Moratorium, signed by the main grain trading companies, also provides a framework for the blocking of soy originating from farms with deforestation after 2008, creating an additional layer of market-based enforcement (Heilmayr et al., 2020). While a step in the right direction, these agreements alone have not been able to eliminate illegal deforestation, especially in the cattle sector. Currently, slaughterhouses only analyze the environmental compliance of their direct suppliers – the farms from which they purchase animals for slaughter – while indirect suppliers earlier in the production chain often remain unmonitored (Brandao Jr, 2023).

The Brazilian cattle supply chain is complex, characterized by multi-stage production systems — from breeding and rearing to fattening—, which hinders traceability and environmental monitoring, particularly for indirect suppliers—who do not sell animals directly to slaughterhouses. Studies indicate that the number of unmonitored indirect suppliers outnumbers direct suppliers by more than three times (Skidmore et al., 2021). Recent evidence shows that 81% of deforestation within supply chains covered by Cattle Agreements occurs among indirect suppliers (Gibbs et al., 2026). As a result, this monitoring failure creates a substantial leakage loophole, allowing non-compliant suppliers to remain active in the supply chain. This dynamic undermines the effectiveness of the Cattle Agreements and weakens the credibility of the sector, ultimately compromising the environmental integrity and international

reputation of Brazil's agribusiness (Rajão et al., 2020; Levy et al., 2023; Vaccarezza Sevilla et al., 2025).

Literature has addressed this issue from various perspectives. Early studies quantified the implications of the Forest Code following its revision in 2012 by developing methods to assess the application of rules using high-definition satellite imagery, as well as mapping the potential of the Environmental Reserve Quota (CRA) as a market-based compensation mechanism provided for in the law (Soares-Filho et al., 2014). Other analyses used geospatial and economic modeling to simulate the impacts the full application of the FC would have in preventing forest loss and conserving biodiversity (Brock et al., 2021; Soterroni et al., 2018). However, many of these studies tended to simplify the complexity of governance and neglected the specific dynamics of supply chains and their interactions with market policies. In particular, there is limited understanding of how environmental agencies and private sector initiatives can effectively support producers in the transition to legal compliance, that is, the potential to drive the environmental regularization of rural properties by fulfilling all Forest Code rules.

The state of Mato Grosso provides a recent example of how technological and regulatory innovation can accelerate the CAR validation process. The state has advanced this process with the implementation of CAR Digital 2.0, a system that reconstructs cadastral data using high-resolution cartographic databases. Following this reconstruction, automatic analyses assess compliance with APP and RL requirements, and landowners are notified of the result so that they can take recovery measures if there is a vegetation deficit (SEMA, 2024). As a result, Mato Grosso significantly improved the efficiency of CAR validation. The state doubled the number of analyses initiated and increased the validation from 11% to more than 19% of its state CAR database in 8 months, even though this technique was applied in only 1/3 of the municipalities (Lopes et al, 2025).

In this context, the present study proposes an in-depth analysis of the interaction between the legal requirements of the Forest Code and complementary incentive mechanisms, including CAR Digital 2.0 and the Green Passport program, which is led by the Mato Grosso Institute of Meat (IMAC). The Green Passport program aims to support environmental regularization and individualized traceability within the cattle supply chain, offering incentives such as accelerated CAR analysis and technical assistance for

farmers with up to 100 animals. (ALMT, 2025).

This study develops a geospatial database to document current property-level compliance with the preservation requirements of the Forest Code. It also provides precise information estimates of areas requiring restoration in Mato Grosso, drawing on the best available databases for hydrography, deforestation, and land use. In addition, this study evaluates the role of properties within the cattle supply chain and estimates the potential impact of environmental regularization under different scenarios such as off-property compensation through the CRA versus on-property restoration. It further investigates how initiatives like the Green Passport program and judicial conciliation mechanisms can support compliance with the law.

I will use certain concepts frequently, and for better understanding, the term and its definition are below:

- Validated: CAR of the individual property that has had its analysis process finalized and validated by the environmental agency, approving the framework of LR areas, APP and areas suitable for alternative use. It may or may not result in a liability for the property.
- Regularization: process that the property must undergo to comply with the FC, including the validation of the CAR and approval of a plan to fulfill the liability, if any.
- Compliant: property that meets the requirements of the FC.

This thesis is structured into chapters that present detailed background about the policy and previous work, detail the mix-methods approach, and analysis results of spatial and statistical data, culminating in final considerations and policy recommendations to strengthen forest governance in the state and across the Amazon.

2. Instruments to Drive Environmental Regularization in the Brazilian Cattle Industry

This section is dedicated to examining the theoretical and empirical context of the implementation of the Brazilian Forest Code in the state of Mato Grosso. We focus on the impact environmental regularization has on the volume of areas requiring restoration for conservation and the existing conserved areas within rural properties that exceed legal requirements. We also highlight the complexities of the conservation-production relationship in the cattle supply chain and the role of market initiatives in environmental governance.

2.1 The Forest Code and Mandatory Conservation Instruments

The Forest Code (Law No. 12,651/2012) is the central regulatory framework that establishes general standards for the protection of native vegetation on rural properties (Luiz, 2022). It provides a set of rules to reconcile environmental protection with agricultural production and restoration of degraded areas (Leme et al., 2025). The sustainable forest management promoted by the Forest Code is vital for Brazil to meet its international commitments regarding climate change mitigation and biodiversity conservation (Brock et al., 2021). The Forest Code imposes the obligation to maintain conservation areas within their properties on all rural owners or possessors (Silva, Pinto, & Scarano, 2021). The two main legal instruments for this purpose are Permanent Preservation Areas and Legal Reserves (Silva et al., 2021).

Permanent Preservation Areas are defined based on geographic attributes and serve to preserve water resources, landscapes, geological stability, biodiversity, to facilitate the gene flow of fauna and flora, and to protect the soil (Leme et al., 2025). These protection strips along watercourses vary from 30 to 500 meters, depending on the width of the water body (Brazil, 2012).

For rural areas deforested up to July 22, 2008 — legally as consolidated areas — the FC permits the continuation of agro-silvopastoral, as well as ecotourism and rural tourism activities in APPs. However, it requires vegetation restoration within reduced buffer widths, which vary depending on the property size, measured in fiscal modules (FM) (Brazil, 2012). This concept of a fiscal module for classifying rural properties as small, medium, or large was defined in Brazil by laws 6.746/1979 and 8.629/1993. In MT, fiscal modules range from 30 to 100 hectares, depending on the municipality (Pacheco et al., 2021). The specific buffer requirements applied to consolidated areas are detailed in the methods chapter. That is the legal classification:

- Small property, between 0 up to 4 fiscal modules;
- Medium property, more than 4, but less than or equal to 15 fiscal modules;
- Large property, more than 15 fiscal modules.

Beyond rivers, the APP regime encompasses the surroundings of natural lakes and lagoons, requiring buffers of 100 meters in rural zones, which can be reduced to 50 meters for water bodies smaller than 20 hectares, and 30 meters in urban areas. Additionally, the law stipulates a minimum protection radius of 50 meters around perennial springs and water eyes, regardless of their topographical situation (Brazil, 2012).

In addition to hydrological protections, the FC establishes APP categories linked to topography and specific ecosystems, aimed at mitigating erosion and safeguarding environmentally sensitive areas. Within this scope, slopes or portions with a steepness greater than 45 degrees are designated as APP's, as are the tops of hills, mounds, mountains, and ridgelines with a minimum height of 100 meters and an average slope exceeding 25 degrees.

The legislation also extends protection to specific phytogeographic formations. These include buffer zones of at least 50 meters from the permanently waterlogged areas in wetlands, as well as full protection of ecosystems such as mangroves and coastal sand dune habitats. Additional APP categories include the edges of tablelands or plateaus which require minimum strips of 100 meters, and all areas located at altitudes above 1,800 meters.

However, for rural areas consolidated prior to July 22, 2008, the FC establishes more flexible

transitional restoration requirements. In such cases, the width of vegetation recovery along water margins is reduced and varies according to property size, measured in fiscal modules (Brazil, 2012). Generally, with the reduction, the requirement varies between 5 and 15 meters for small properties (up to 4 fiscal modules), regardless of the river's width. Detailed information is available in the methods chapter.

The Legal Reserve (LR) is a portion of a rural property that must be maintained under native vegetation cover, with required percentage varying according to the biome and vegetation type. According to Article 12 of the FC, the LR serves to ensure the sustainable economic use of the natural resources, of the rural support, the conservation and restoration of ecological processes, and promote biodiversity conservation, while also providing habitat for wildlife and native flora (Brazil, 2012). The proportion of land designated as LR is defined as follows: 80% in forest areas within the Amazon biome, 35% in Cerrado areas located within the Legal Amazon, and 20% in grasslands and Cerrado areas outside the Legal Amazon (Brazil, 2012). The Legal Amazon is an administrative region established by Complementary Law 124/2007, encompassing the states of Acre, Amapá, Amazonas, Mato Grosso, Rondônia, Roraima, Tocantins, Pará, and part of Maranhão (Brazil, 2007).

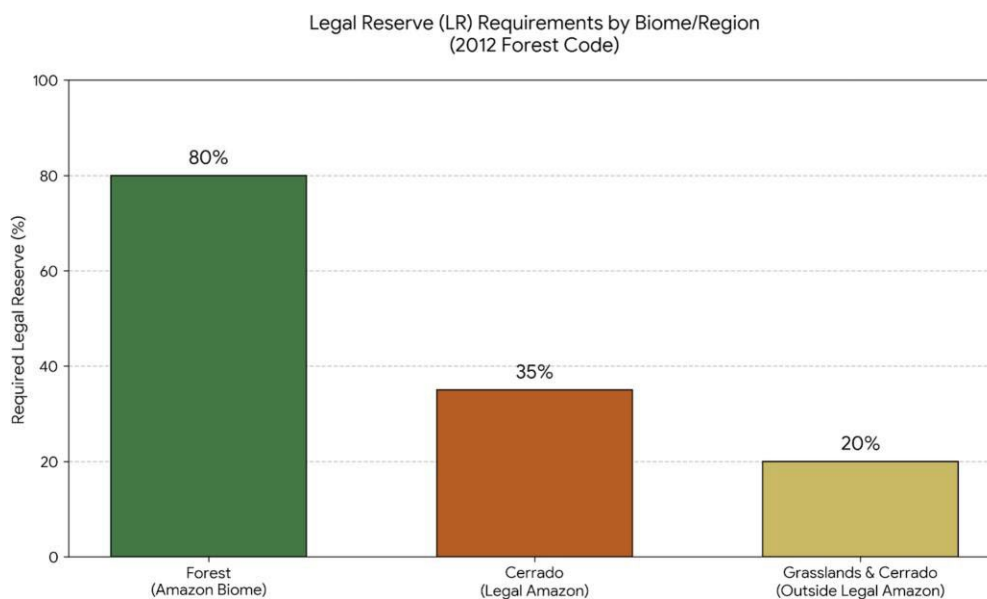


Figure 1: Legal Reserve vegetation currently required by the Brazilian Forest Code.

The concept of consolidated areas, introduced by the 2012 FC, also applies to LRs, granting

specific provisions for areas deforested prior to July 22, 2008. For properties in this situation, the law allows owners with an LR deficit—defined as the difference between the existing vegetation and the legally required percentage—to compensate for this liability through alternative mechanisms rather than on-site restoration.

One key mechanism is the purchase of Environmental Reserve Quotas (CRA), which can be acquired from properties with LR surplus within the same biome. Compensation may also occur through the purchase of land with surplus LR or through the donation of areas located within public Conservation Units that are pending land regularization (Brazil, 2012).

In addition, the FC provides differentiated treatment for small properties (up to four fiscal modules). For these properties, liabilities associated with deforestation prior to July 22, 2008 are effectively regularized through amnesty provisions, allowing the LR to consist of the native vegetation existing on that time, even if below the legal percentage, without the obligation for restoration or compensation (Brazil, 2012).

Furthermore, Article 68 of the 2012 FC guarantees the maintenance of acquired rights under the previous legal frameworks. Under this provision, landowners or possessors who cleared native vegetation in compliance with the legislation in force at that time are exempt from restoring or compensating to meet current LR requirements (Brazil, 2012). This exemption applies to deforestation carried out under earlier regulations, including the old versions of the FC and subsequent Provisional Measures issued between 1996 and 2012, which carry the force of law in Brazil. Notably, the 1965 Code established LR requirements of 50% for the Amazon and 20% for the Cerrado (Brasil, 1965). These thresholds were later increased in May 26, 2000 to 80% in the Forest and 35% in Cerrado areas within the Legal Amazon, maintaining 20% for other biomes under Provisional Measure No. 1,956-50/2000 (Brasil, 2000). Since then, these thresholds have remained unchanged and were consolidated in the 2012 Forest Code.

This is the interpretation that the State of Mato Grosso (MT) makes regarding the numerous Provisional Measures that were issued between 1996 and 2000 by the President of the Republic and that is in State Decree MT No. 1,031/2017, which regulates the application of CAR rules in MT (Mato Grosso, 2017).

However, there is legal discussion on the subject, since Provisional Measure 1,511, of July 25, 1996, although it maintained the LR rates at 50% in the Amazon, restricted clear-cutting to only 20% of properties in forest phytophysiognomies (vegetation type in biome), which would consequently increase protection to 80% in the Forest (Brazil, 1996). The problem is that, in Brazilian legislation, the text that effectively increased the LR was modified by Provisional Measure No. 1,956-50/2000.

Since this work is limited to the analysis of the CAR in the State of MT, we will apply the understanding of State Decree MT No. 1,031/2017, which is also followed by the State Court of Justice (TJMT, 2025). In MT, only the requirements for Forest and Cerrado phytophysiognomies apply, as it is part of the Legal Amazon (Brazil, 2007).

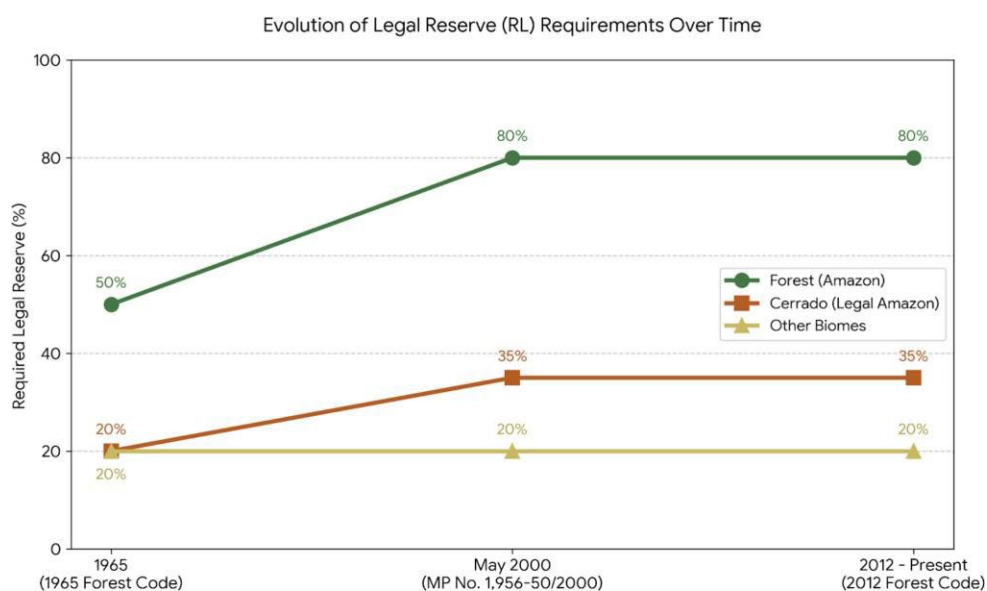


Figure 2: Evolution of legal reserve indices.

Landowners who complied with the LR percentages established under previous legislation are exempt from restoring the LR to the level indicated in the 2012 FC. For example, properties that maintained 50% LR in the Forest biome and 20% in the Cerrado prior to the regulatory changes in 2000 are not required to restore the areas to the current thresholds for 80% or 35%, respectively. However, this exemption does not grant them the right to deforest more area until reaching the lower historical percentages. In other words, it preserves past legal compliance but

does not authorize additional clearing, according to article 68 of the FC and State Decree 1,031/2017.

Another rule that applies to LR is that provided for in Article 15 of the Forest Code, which allows the inclusion of APP in the calculation of LR under certain conditions, such as the area being conserved or in the process of recovery and registered in the CAR. After the CAR is validated, if there is a deficit of APP, the state agency requires the recovery of the liability. As a consequence, this rule ends up being applied automatically in the CAR validation process.

The implementation of the FC is primarily operationalized through the CAR, which is a self-declaratory electronic public record that integrates environmental information of properties into a georeferenced map (Leme et al., 2025). The spatial information declared by the rural owner undergoes analysis and validation by the state environmental agency, which may approve or reject the location of the LR and APP. In the event of an APP or LR deficit, the farmer must enroll in the Environmental Regularization Program (PRA), which establishes the necessary measures to obtain legal compliance, such as restoration or compensation of liabilities (Pacheco et al., 2021), (Luiz, 2022). For LR restoration, the rural owner has an extensive period of up to 20 years, provided they restore at least 5% of the deficit annually (Brazil, 2012). This recovery process usually begins after the CAR is validated, at which point rural landowners have clarity about the area they can exploit and the area they must recover within their property (Carvalho et al., 2024).

Despite being a crucial instrument for conservation, the validation of the CAR is a major institutional challenge. After more than a decade of the law being in effect in MT the majority of registrations have yet to be validated (Carvalho, Rausch, Gibbs et al., 2024). A study conducted with data up to August 2025 showed that Mato Grosso ranked third nationally and first in the Legal Amazon in the highest number of CAR validations, with just 19% of CARs validated, more than the national average (6%) (Leme et al., 2025). The complexity of applying FC standards hinders CAR validation (Leme et al., 2025), delaying environmental restoration and complicating control over illegal deforestation.

2.2 Environmental Compensation Modalities and the Environmental Reserve Quota Market

The Brazilian Forest Code (Law No. 12,651/2012) instituted environmental compensation as a flexible mechanism for the regularization of Legal Reserve liabilities whose deforestation occurred in a consolidated manner prior to the milestone of July 22, 2008 (Brazil, 2012). To achieve legal compliance without the necessity of removing productive areas from operation, federal legislation stipulates four distinct modalities for off-property compensation, which must obligatorily occur within the same biome as the deficit area (Luiz, 2022). Rural producers may compensate by acquiring Environmental Reserve Quotas, leasing environmental servitudes, donating private lands within pending public Conservation Units, or registering equivalent surplus properties (Brazil, 2012).

Within the state of Mato Grosso, this federal directive was regulated by State Decree No. 1,757, of November 24, 2025, which established the State Policy for Legal Reserve Compensation (Mato Grosso, 2025). The state regulation ratifies the four compensation modalities of the Forest Code and provides for the creation of the State Environmental Reserve Quota, which is subject to specific technical regulations, establishing the SIMCAR Compensation computerized system to operationalize the compensations (Mato Grosso, 2025). This electronic system aims to manage the supply and demand of eligible areas in a fully digital manner, requiring that both the debtor property and the ceding property possess a fully validated Rural Environmental Registry by the state environmental agency prior to any transaction (Mato Grosso, 2025). The Mato Grosso decree also categorizes priority areas for compensation, seeking to favor the recovery of watersheds and the conservation of fragile ecosystems mapped by socioeconomic and ecological zoning. Finally, it establishes the specificity of allowing compensation in Conservation Units of other states of the federation (Mato Grosso, 2025). For this interstate compensation to occur, the area is required to be located within the same biome, conditioned upon the prior execution of a formal cooperation agreement and the compatibility of the computerized systems between the involved states (Mato Grosso, 2025).

When contrasting the Mato Grosso state regulation with Federal Decree No. 9,640, of De-

cember 27, 2018, differences in scope are observed that complement each other in the practice of environmental governance (Brasil, 2018). While the state legislation disciplines the procedural, territorial, and land tenure rites for all forms of physical compensation within its borders, the federal decree acts strictly in regulating the Environmental Reserve Quota as a national economic and financial asset (Brasil, 2018). Federal regulation requires that quota registration be carried out on commodity exchanges or financial settlement systems previously authorized by the Central Bank of Brazil, in order to provide backing and market security to forest preservation (Brasil, 2018). Both regulations share the fundamental similarity of requiring equivalent size and biome for the exclusive regularization of consolidated liabilities (Brasil, 2018; Mato Grosso, 2025). The interconnection between the state and federal spheres occurs compulsorily through the data management infrastructure, since the state system SIMCAR Compensation will need to operate in a perfectly integrated manner with the national module of the Rural Environmental Registry System to guarantee the validity of the generated titles, prevent the double counting of areas, and enable liquidity in financial markets (Brasil, 2018; Mato Grosso, 2025).

This normative structuring serves as the foundation for the development of the Environmental Reserve Quota market, which presents itself as one of the most promising opportunities for reconciling the agricultural economy and conservation (Rajão, 2015). This market possesses the capacity to drastically reduce regularization costs for producers with deficits, given that conserving surplus native areas through the acquisition of quotas is considerably less onerous than active ecological restoration of degraded areas on lands with high agricultural aptitude (Rajão, 2015; Bernasconi, 2013). Simultaneously, the commercialization of quotas acts as a veritable payment for environmental services mechanism, attributing tangible economic value to the standing forest and incentivizing property owners to maintain their native vegetation surpluses beyond legal requirements (Bernasconi, 2013; Silva & Ranieri, 2014). The success of this environmental market and the consequent liquidity of the assets depend directly on the State's agility in validating rural registries and the effective inclusion of small properties in the compensation system, democratizing access to the benefits of the green economy (Oliveira, Miotto, & Macedo, 2024).

However, if there are significant difficulties or excessive costs involved in registering surplus vegetation as a financial asset, there will be resistance to entering the Environmental Reserve Quota market, so producers will end up opting for other compensation mechanisms that may be easier to implement at the state level, such as environmental easements or the acquisition of parcels of land in protected areas for environmental regularization.

2.3 Challenges in the Interpretation and Practical Application of the Forest Code within the Rural Environmental Registry

The implementation process of the Brazilian environmental policy reveals a profound gap between the theoretical formulation of the Forest Code norms and their practical application in the design of the Rural Environmental Registry of properties. The legislation establishes a complex framework of rules for the delimitation of Permanent Preservation Areas and Legal Reserves, the translation of which into spatial polygons requires advanced knowledge in geoprocessing and remote sensing, as well as access to geographic databases with a high level of detail (Mariano et al., 2024). This technical requirement disproportionately affects small rural producers, who frequently lack financial resources and adequate access to technical assistance and rural extension to properly register their properties (Lopes et al., 2024). Faced with this technical vulnerability, many declarations are filled out incorrectly, resulting in inconsistent data that stall the analysis process and prevents these family farmers from accessing the benefits of environmental regularization, as they are unable to accurately map their consolidated use areas and their environmental liabilities (CNMP, 2022).

Additionally, the normative complexity is exacerbated by the fact that several rules depend directly on the interpretation and regulation by state environmental agencies. A paradigmatic example of this interpretative subjectivity is the application of the fallow rule, known as “pousio”. The Forest Code defines fallow land as the temporary interruption of agroforestry activities for a maximum period of five years to allow for the recovery of the soil's capacity for use or physical structure (Brazil, 2012). However, the practical application of this definition in mapping requires rigorous criteria. In the state of Mato Grosso, the State Secretariat for the

Environment interprets that any area of secondary vegetation that has been in continuous regeneration for a period exceeding five years loses its status as a consolidated use area, and must obligatorily be mapped and protected as a Native Vegetation Area (SEMA, 2024). This temporal and ecological transition forces public administration and producers to conduct meticulous analyses of extensive historical series of satellite images, frequently generating disagreements and legal uncertainty regarding the actual classification of features when quantifying the area subject to restoration (CNMP, 2022).

Another structural obstacle to the enforcement of the law is the enormous difficulty of creating and maintaining high-resolution geographic databases across a territory of continental dimensions. The accurate validation of property boundaries, hydrography, and forest remnants demands cartographic inputs at highly detailed scales, which are costly and technically complex to produce uniformly across different biomes (CNMP, 2022). The absence or obsolescence of these official reference bases results in the generation of spurious polygons and cartographic noise during automated spatial intersections, primarily due to the difference in spatial resolution between historical satellite images used to prove land use at the two thousand and eight legal milestone and the modern, very high-resolution images employed in mapping current forest cover (SEMA, 2024).

Consequently, the validation process of the Rural Environmental Registry faces severe operational bottlenecks that delay public policy throughout the country. The low quality of self-declared data requires environmental agencies to perform multiple analysis cycles and issue notifications requesting rectifications and clarifications from registrants. Nevertheless, states encounter the chronic lack of response from rural producers to the requirements issued by the Environment Agencies to adjust the property's registry (Lopes et al., 2024). Communication difficulties are exacerbated by the lack of internet access in remote areas or by the fact that the contact information belongs to third-party consultants who registered the producer years ago and with whom the producer has lost contact, causing the process to stall and the producer often only becomes aware of the pending issue when facing commercial barriers to the sale of their production (Lopes et al., 2024).

The difficulty in interpreting and applying the law extends to other critical issues of environmental regularization, such as the identification of frauds related to land grabbing, the intentional fragmentation of properties to circumvent restoration rules linked to property size, and the challenge of distinguishing illegal deforestation from duly authorized suppression areas. The self-declaratory nature of the system, coupled with the initial lack of integration with the land tenure bases of land institutes, allowed the system to be improperly used to register overlapping claims on Indigenous Lands and public Conservation Units (CNMP, 2022; Chiavari et al., 2023). All these factors demonstrate that the flexibilities intended by the legislation created a fragmented regulatory environment that demands an unprecedented technological, analytical, and human infrastructure from the State to ensure that rules translate into effective territorial governance and conservation (Chiavari et al., 2023; Rajão et al., 2020).

With the aim of applying the rules of the Forest Code and contributing to the discussion on environmental regularization mechanisms, this work will apply spatial and statistical data analysis, using available geographic map databases, such as deforestation, hydrography, land use type, among others, to create a geodatabase that provides the most accurate spatial information to date on compliance with the Forest Code requirements for conservation and restoration at the property level in Mato Grosso. I will also analyze the impact of regularization in different scenarios and the use of CRA (Environmental Reserve Quotas) for compensation and sectoral incentive instruments, such as the Green Passport program in the livestock sector. I will apply rules that previous works have not yet done, such as the reduction of the Legal Reserve by Provisional Measure between April and May 2000, in addition to using more recent data and hydrography to calculate the compliance of Permanent Preservation Areas (APP) in properties with consolidated areas.

In light of the methodological and normative complexities inherent in environmental regularization, it is fundamental to emphasize that the present work does not intend to replace the official procedures and analysis workflows conducted by state environmental agencies for the validation of the Rural Environmental Registry. The scope of this research focuses on providing an updated geospatial diagnosis to enhance the understanding of forest governance and

environmental compliance in the livestock supply chain in the state of Mato Grosso, acting as a complementary effort to public management. Similarly, any discrepancies in the results presented here do not indicate that previous models developed by other researchers in the past are incorrect. The spatial modeling of the Forest Code is an intrinsically dynamic scientific process, in which each empirical study is strictly guided and delimited by the quality, spatial resolution, and availability of existing geospatial databases and information at the exact moment of its elaboration (Rajao et al., 2020). The constant advancement of geotechnologies and the creation of new high-precision cartographic databases allow for progressive refinements, which add to the legacy of previous research efforts for the effective understanding and implementation of environmental policy in Brazilian territory.

2.4 The Cattle Supply Chain and Monitoring Challenges

Cattle ranching is the main driver of pressure on native vegetation in Brazil, occupying most of the deforested area in the Amazon (Brandão Jr. et al., 2023). In this supply chain, cattle typically pass through multiple properties (calving, rearing, fattening) before reaching the slaughterhouse (Skidmore et al., 2021; Rajão et al., 2020).

Seeking to curb illegal deforestation in the Amazon, since 2009 the Brazilian Federal Prosecutor's Office has been conducting an analysis of the environmental compliance of farms that supply animals to slaughterhouses. Upon finding that a company acquired cattle from non-compliant rural properties, it pressured slaughterhouses operating in that biome to sign a legal instrument called a Conduct Adjustment Agreement, obligating them to stop buying from properties when any of the following criteria are identified: the existence of illegal deforestation (without prior authorization from environmental agencies, detected by PRODES); overlapping farm areas with conservation units or indigenous lands; and if the property owner is on the "dirty list" of modern-day slave labor. Due to pressure from Greenpeace, the three largest slaughterhouses in Brazil (JBS, Marfrig, and Minerva) also signed another agreement, called the Public Commitment of Livestock Farming, focused on analyzing deforestation in the Amazon post-July 2008.

To prove compliance, slaughterhouses undergo independent audits, with results sent to Federal Prosecutors. If purchases from irregular properties are confirmed, companies must adjust their monitoring processes and may face fines per animal that originated in disagreement with the rules. The application of these rules was unified in 2020 through the Monitoring Protocol for Cattle Suppliers in the Amazon (MPF, 2024). Slaughterhouses that signed the Conduct Adjustment Agreement and underwent auditing showed only 4% irregularities in their Amazon operations in 2025 (Imaflora, 2025).

To verify compliance with the agreements, slaughterhouses are subject to independent audits, with the results forwarded to the Federal Prosecutor's Office. If the purchase of animals from irregular properties is found, companies must adjust their monitoring and blocking processes for non-compliant farms to prevent recurrence of the error and may have to pay a fine per animal originating in violation of the rules.

The application of the rules of these agreements was unified from 2020 onwards, through a document called the Amazon Cattle Supplier Monitoring Protocol, as well as the independent audit process to assess compliance with the criteria by slaughterhouses (MPF, 2024). Slaughterhouses that signed the Conduct Adjustment Agreement with the Federal Prosecutor's Office and carried out the audit process presented only 4% irregularities in their operations in the Amazon in 2025 (Imaflora, 2025).

These cattle sector agreements represented a paradigm shift in environmental governance by holding slaughterhouses responsible for the environmental liabilities of their suppliers and were crucial in reducing the purchase of cattle from farms with recent deforestation (Gibbs et al., 2016). However, the effectiveness of this command-and-control instrument faces significant structural and technical obstacles that limit the capacity of the Federal Prosecutor's Office to ensure full compliance (Barreto & Brito, 2023).

The main challenge faced by the Federal Public Prosecutor's Office and the signatories of the agreements lies in the invisibility of indirect suppliers, who constitute the breeding and rearing links in the livestock chain (Skidmore et al., 2024). The current monitoring system focuses almost exclusively on direct suppliers—the fattening farms that sell cattle for slaughter—leaving

a vast network of upstream farms outside of socio-environmental scrutiny (Alix-Garcia & Gibbs, 2017). This monitoring gap facilitates the practice known as "cattle laundering" or triangulation, in which animals raised in embargoed or illegally deforested areas are transferred to a "clean" (compliant) property before the final sale to the slaughterhouse, thus circumventing the blocking filters established by the agreement (Gibbs et al., 2016; Rajão et al., 2020). Studies indicate that most illegal deforestation in the beef supply chain occurs precisely among unmonitored indirect suppliers, which compromises the overall effectiveness of the agreement in reducing regional deforestation (Skidmore et al., 2021). In Mato Grosso and Pará, it is estimated that 34% of exported cattle may be contaminated by potentially illegal deforestation originating from indirect suppliers (Rajão et al., 2020).

Logistical complexity and a lack of integration of government data hinder enforcement against this triangulation. Although the Animal Transit Guide records the movement of animal batches for sanitary reasons and could serve as a tool to track indirect suppliers, access to this data is frequently restricted or not transparently integrated with the Rural Environmental Registry and PRODES deforestation data (Barreto & Brito, 2023; Brandão Jr. et al., 2023). The absence of a mandatory national individual traceability system, from the animal's birth to its slaughter, prevents the Federal Public Prosecutor's Office and independent auditors from accurately verifying the origin of each slaughtered animal, maintaining dependence on documentary systems that are susceptible to fraud (Vaccarezza Sevilla et al., 2025).

Another challenge is market leakage, resulting from the incomplete coverage of the Conduct Adjustment Agreement. Although the agreements cover a significant portion of the slaughter capacity in the Amazon, especially from slaughterhouses with Federal Inspection focused on export, there is still a market formed by non-signatory slaughterhouses, which mainly serve domestic consumption and operate with fewer environmental restrictions (Alix-Garcia & Gibbs, 2017; Brandão Jr. et al., 2023). In the last audit cycle of the agreement, slaughterhouses that represented 74% of the slaughter in Mato Grosso participated, therefore there is a leakage of 36% in this State (AdT, 2025). Producers blocked by the slaughterhouses that are signatories to the Conduct Adjustment Agreement find outlets for their production in these unmonitored plants,

diluting the deterrent impact of the agreement (Alix-Garcia & Gibbs, 2017). A 95% spatial overlap between the purchasing zones of slaughterhouses with and without the Conduct Adjustment Agreement aggravates this scenario, creating unequal competition where deforestation continues to be financed by actors outside the agreement (Brandão Jr. et al., 2023).

Furthermore, monitoring compliance with the agreements faces technical difficulties related to the standardization of audits. Historically, there have been discrepancies in the criteria and parameters used by geomonitoring companies contracted by meatpacking plants to verify supplier compliance, which has generated asymmetries in the results presented to the Federal Public Prosecutor's Office (Guimarães, Stabile, & Moutinho, 2023). In response, initiatives such as the "Boi na Linha" program sought to harmonize monitoring and audit protocols, unifying the rules to provide greater legal certainty and transparency to the results (Guimarães, Stabile, & Moutinho, 2023). The last two audit cycles (years 2024 and 2025) were carried out using this standardized protocol (Imaflora, 2025).

Finally, the monitoring of the agreements occurs within a context of fragility in the validation of land and environmental data. The Rural Environmental Registry, the basis for verifying compliance, is self-declared and presents overlapping rates with indigenous lands and conservation units, in addition to significant delays in analysis and validation by state agencies (Nunes et al., 2024; Carvalho et al., 2024). This precariousness of public data imposes on the Federal Public Prosecutor's Office the challenge of monitoring private agreements based on state information that often does not reflect the land reality, requiring continuous efforts to improve databases and integrate territorial intelligence systems (Nunes et al., 2024).

In February 2026, the Federal Prosecutor's Office informed signatory slaughterhouses that, starting in 2027, an analysis of the environmental compliance of level 1 indirect suppliers will be required; that is, those who sell animals to the companies' direct suppliers. It also informed them that other levels will also be subject to analysis in the future (MPF, 2026). This measure is essential to expand compliance analyses to the entire chain, pressuring rural producers to comply with the Forest Code.

2.5 The Green Passport Program: A Complementary Instrument

Given the limitations of public monitoring, the Green Passport Program emerged as a private initiative by the Mato Grosso Meat Institute, enacted as State Law No. 11,153/2025. The program aims to verify socio-environmental compliance and promote sustainable development by assisting in property regularization. The project focuses on the individual traceability of animals and linking it to the socio-environmental compliance status of farms (ALMT, 2025).

The Green Passport requires adherence to the Reinsertion and Monitoring Program for properties with illegal deforestation after 2008 (IMAC, 2025). The program obliges the producer to isolate and regenerate illegally deforested areas under continuous monitoring and requires the payment of a compensatory fine per hectare of forest removed without authorization from state agencies (IMAC, 2025). By fulfilling these steps, the farm is considered eligible, receiving a Certificate of Socio-environmental Compliance, allowing them to sell animals again to slaughterhouses that are signatories to cattle agreements.

A key benefit for livestock farmers is the preferential analysis of the Rural Environmental Registry by the state environmental agency (ALMT, 2025). This prioritization is a significant incentive, as it mitigates the bottleneck of slow validation of registrations (Carvalho et al., 2024). Participation in the Green Passport is voluntary, but becomes mandatory for properties with illegal deforestation, which will have up to 48 months to join the program, carrying out individual animal identification and initiating the environmental regularization process (ALMT, 2025).

2.6 Geospatial Data Integration for Compliance Analysis in Cattle Chain

Accurate diagnosis of legal compliance in Mato Grosso requires the integration of geospatial databases to overcome the heterogeneity of the Forest Code regulations and the shortcomings in monitoring systems (Carvalho et al., 2024; Rajão et al., 2020).

With the aim of contributing to the verification of the compliance of rural properties with all the rules of the Forest Code, we developed a framework using the best available databases. Afterwards, this will be combined with information from the Animal Transport Guide, a document that records

the movement of animals in lots between rural properties, following the methodology of Brandão Jr. et al. (2023). This process allows:

- a. Mapping properties with environmental problems, identifying and quantifying the deficit of Legal Reserves and Permanent Preservation Areas in rural properties within the beef supply chain.
- b. Identify properties with a quantity of native vegetation exceeding the requirements of the Forest Code, which are potential environmental assets for the issuance of Environmental Reserve Quotas (Oliveira et al., 2024; Carvalho et al., 2024).
- c. Identify the role of each property in the beef supply chain (direct or indirect supplier) and quantify the impact of regularization on the chain, obtaining estimates of properties that can benefit from programs such as the Green Passport for environmental regularization.

This analysis will provide estimates on how regularization will influence land use and forest conservation in different scenarios in Mato Grosso.

3. Methodology

Our study area is the state of Mato Grosso, Brazil's main agricultural powerhouse. Mato Grosso stands out simultaneously as the largest national producer of soybeans and the holder of the largest cattle herd in the country (Arvor et al., 2012; Vale et al., 2019). If it were a country, Mato Grosso would rank as the world's sixth largest cattle producer and the fourth largest soybean producer, ahead of countries such as China and Australia (Macedo et al., 2012). Mato Grosso is covered by three ecological biomes, the Amazon, Cerrado, and Pantanal. Here, we assessed Forest Code compliance on rural properties in the state throughout all three biomes.



Figure 3: Study area - Mato Grosso, Brazil.

3.1 Research Design and Database

This study applies quantitative modeling and spatial data analysis to questions of environmental

governance. We estimated compliance with the Brazilian Forest Code in Mato Grosso at multiple scales, including property, municipality, and state levels.

We developed a method to automate and audit the calculation of the environmental compliance balance, including the deficit and surplus of Legal Reserve and the deficit of Permanent Preservation Areas, in rural properties, by parameterizing Forest Code regulations. In cases of deficit, it was then classified whether they are eligible for compensation or restoration within the property.

We implemented a four-stage sequential geospatial vector processing architecture, applied to the full population of rural property polygons registered in Mato Grosso Rural Environmental Registry System, available in SEMA Geoportal. All procedures were executed in the ESRI:102033 coordinate reference system, South America Albers Equal Area Conic, a standard projection for continental-scale quantitative area analysis in South American contexts. To accelerate processing, all geometries were stored in GeoParquet format, and spatial overlap operations utilized Sort-Tile-Recursive tree indexing, known as STRtree, to maintain intersection queries tractable in datasets encompassing millions of polygons.

To perform the analysis, secondary data processing from the rural registry was used, specifically from the Mato Grosso Rural Environmental Registry System. Other databases were also used, as described in Table 1.

Table 1: Database sources.

Layer	Source	Format and period (year)
Land use and land cover (PRODES)	INPE	Vector (1988 to 2024)
Hydrography of rivers and lakes	SFB	Vector (2023)
Forest typology	RADAM Brasil	Vector (1970)
Map of remaining vegetation (2008)	Geoportal (SEMA-MT)	Vector (2008)
Property boundaries and registry data	Geoportal (SEMA-MT)	Vector (2026)
Map of validated registry	Geoportal (SEMA-MT)	Vector (2026)
Map of Digital Registry	Geoportal (SEMA-MT)	Vector (2026)
Hydrology (rivers, lakes, springs)	Geoportal (SEMA-MT)	Vector (2026)
Secondary vegetation	INPE	Vector (2008 and 2012)
Protected territories	INPE	Vector (2026)
Settlements	INCRA/INTERMAT	Vector (2026)

The tabular database encompassed descriptive variables of the properties, such as total declared area, size in Fiscal Modules (FM), the percentage of biomes at the property level (Forest and Cerrado, based on RADAMBRASIL mapping, as applied by SEMA-MT), and the history of

deforestation extracted from geospatial intersections of the Amazon Deforestation Monitoring Project (PRODES) maintained by Brazil's National Institute for Space Research, with vegetation cover mappings referring to the temporal milestones of 2000, 2008, and current days.

To enable analyses regarding the application of LR indices under the previous legislation and the Forest Code exceptions for consolidated areas (deforested by 2008), it was necessary to develop two distinct consolidated areas, both using PRODES data. One consolidated area encompassing the cumulative deforestation up to the year 2000 and another up to the year 2008. To build the remaining vegetation map, all polygons of native vegetation suppression (PRODES/INPE), available until 2024, were merged and dissolved. Then, secondary forest data from INPE, covering the year 2022, were added to the vegetation map from PRODES to represent the remaining primary and secondary vegetation. Next, bodies of water (rivers and lakes) were removed.

The inclusion of the secondary vegetation map aligns with the fallow rules (*pousio*), which is the interruption of agricultural activities or uses to allow the recovery of the soil's use capacity or physical structure (Brazil, 2012). However, after a period of 5 years without land use and with clear regeneration, the state environmental agency understands that secondary vegetation becomes legally protected, becoming part of the vegetation base (SEMA, 2024).

Overlaps with special territories were calculated by the geometric intersection of each CAR polygon with four reference datasets from official national sources. The first consists of demarcated Indigenous Lands. The second comprises Conservation Units, excluding Environmental Protection Areas, which allow economic land use and do not impose the same deforestation restrictions as strictly protected and restricted-use conservation categories. The third dataset includes Land Reform Settlement Projects identified and georeferenced by the National Institute for Colonization and Agrarian Reform (INCRA) and the Mato Grosso Land Institute (INTERMAT). The fourth covers Quilombola Territories recognized by the Palmares Cultural Foundation. For each property, we recorded the presence or absence of overlap with each category, the area of overlap in hectares, and the percentage relative to the property's total area.

3.2 Computational Tools and R Processing

The development of the calculation algorithm and the structuring of batch processing scripts were carried out in the R programming environment. The computational environment was optimized for Big Data processing through the dplyr libraries for dataframe manipulation and variable transformation, and the parquet format (via arrow package) to ensure efficiency in reading and writing matrices containing hundreds of thousands of records.

The code architecture was designed in a logical sequence (pipeline) that translated legal hermeneutics into mathematical and boolean logic operators, eliminating interpretative biases and ensuring uniform application of the law, regardless of the geometric scale of the property.

3.3 Logical-Mathematical Parameterization of the Legal Reserve

The analytical core of the model focused on the determination of the Base Legal Reserve Requirement and its consequent ramifications for restoration, compensation, or surplus consolidation obligations.

3.3.1 Nominal Requirements, Legal Ceilings and Floors

The gross Legal Reserve requirement was initially calculated based on Article 12 of the FC, weighting the territorial proportion of biomes occurring on the property (Forest and/or Cerrado). For each property, a historical Ceiling (C) and Floor (F) requirement were defined, formulated according to the protection percentages required in the Amazon and Cerrado biomes:

$$C = A_{\text{total}} \times (0.80 \cdot F_{\text{forest}} + 0.35 \cdot F_{\text{cerrado}})$$

$$F = A_{\text{total}} \times (0.50 \cdot F_{\text{forest}} + 0.20 \cdot F_{\text{cerrado}})$$

Where A_{total} represents the property area and F the proportion of the respective biome.

3.3.2 Temporal Milestones and Transition Rules (Arts. 67 and 68)

The stage of greatest computational complexity consisted in applying amnesty or consolidation of past use to the properties, avoiding the occurrence of undue charges or illegal exemptions of recent liabilities. The method established the remaining vegetation (V) in the milestone years by subtracting the Alternative Land Use (ALU) areas from the total property area:

$$V_{2000} = \max(0, A_{\text{total}} - ALU_{2000})$$

$$V_{2008} = \max(0, A_{\text{total}} - ALU_{2008})$$

Then, the script applied the property size condition to define the Base Requirement (E_{base}):

- **Small Properties (Fiscal Modules ≤ 4):** Under the aegis of Art. 67, the requirement was mathematically limited to the vegetation existing on July 22, 2008:

$$E_{\text{base}} = \min(C, V_{2008}).$$

- **Medium and Large Properties (Fiscal Modules > 4):** Under Art. 68, properties that conducted suppression respecting the legislation in force at the time (cutoff in the year 2000) were exempted from recomposition beyond the remaining vegetation of that year, provided that the historical floor of the previous law was respected in all subsequent years (50% forest / 20% cerrado):

$$E_{\text{base}} = \max(F, \min(C, V_{2000})).$$

3.3.3 Mitigation of Double Limitation in Mixed Biomes

A methodological advance proposed by the script was the treatment of properties with mixed phytophysionomies (vegetation type in the biome). Instead of applying the consolidated limits in isolation by biome — which generated a systemic overtaxation effect in the original database —, the script determined the temporal limitation on the total of the property (E_{base}) and, at a later time, distributed such requirement proportionally to the representation of each biome in the property. This technique allowed calculating not only the deficit by biome type in the property,

but also the surplus, important information for calculating the availability of surpluses for compensation.

3.4 Calculation of Liabilities (Restoration and Compensation) and Surplus

Once E_{base} was defined, the model proceeded to qualify the current condition of the property in relation to its contemporary native vegetation cover (V_{actual}). The modeling followed the algorithmic structure below:

Total Deficit (D_{total}): The metric that expresses the divergence between the legal requirement for vegetation and the current reality.

$$D_{\text{total}} = \max(0, E_{\text{base}} - V_{\text{actual}})$$

In situ Restoration (R_{insitu}): Liability generated by illicit deforestation after the July 22, 2008 milestone (ALU_{post2008}). Any deficit resulting from this recent suppression mandates local restoration.

$$R_{\text{insitu}} = \min(D_{\text{total}}, ALU_{\text{post2008}})$$

Liability Compensation (C_{cra}): The balance of the deficit generated by suppressions prior to 2008, which becomes subject to regularization through off-site compensation mechanisms (such as Environmental Reserve Credits).

$$C_{\text{cra}} = \max(0, D_{\text{total}} - R_{\text{insitu}})$$

Legal Reserve Surplus (S_{rl}): For properties where preservation exceeds the determination of the law, the balance of environmental assets (Surplus) was quantified.

$$S_{\text{rl}} = \max(0, V_{\text{actual}} - E_{\text{base}})$$

3.5 Modeling of Permanent Preservation Areas

Permanent Protection Areas were delimited following the rules of Art. 4 of the FC, using the

vector hydrographic dataset published by the Brazilian Forest Service as an official spatial reference for CAR compliance. The riparian strip widths were assigned according to the watercourse width: 30 m for rivers up to 10 m wide; 50 m for rivers between 10 m and 50 m; 100 m for rivers between 50 m and 200 m; 200 m for rivers between 200 m and 600 m; and 500 m for rivers more than 600 m high. For natural lakes and lagoons in rural areas, the width of the protection zones varies from 30 m for bodies of water with less than 1 ha to 100 m for those with more than 20 ha.

The buffers were calculated based on linear and polygonal hydrographic geometries, with dissolution of overlap zones, so that each point on the surface contributes only once to the total Permanent Protection Area and adjacency between watercourses does not inflate the calculated area. The buffer layer was then clipped to each CAR property boundary by geometric intersection, resulting in the intra-property APP layer.

Complementarily, the algorithm processed the adequacy of the APP. For properties with consolidated liability in APP until July 2008, the 'Escadinha Rule' (Staircase Rule) (Art. 61-B) was applied, reducing the required protection size per fiscal module (FM), as FC rule:

- Up to 1 FM: 5 meters (regardless of river width).
- From 1 to 2 FM: 8 meters (regardless of river width).
- From 2 to 4 FM: 15 meters (regardless of river width).
- Above 4 FM: from 20 to 100 meters (proportional to the width of the watercourse, with a minimum of 20 meters for rivers up to 10 meters wide).

The code incorporated conditional triggers contained in the "staircase rule" that limited the mandatory restoration of the APP to percentage strips of the property (10% for properties up to 2 FM; 20% for properties from 2 to 4 FM), safeguarding the legal principle of economic viability of small properties. Deforestation in APP after 2008 was not subjected to these reducers, with its full recomposition being required.

The model we developed does not incorporate APP associated with hilltops, slopes greater than 45 degrees, riparian forest formations known as veredas or coastal restingas, because nationally standardized vector datasets for these features are not available at a resolution compatible with the necessary analytical scale. It is worth noting in this discussion that this methodological limitation is not an exclusive failure of our model, but reflects a persistent technological challenge that affects the governance of environmental agencies themselves today. As a practical example, the environmental agency of the State of Mato Grosso formally recognizes that the identification of terrain APPs, such as slopes greater than forty-five degrees, plateau edges and hilltops, has not yet been included automatically in the scope of CAR Digital analysis (SEMA, 2024).

3.5.1 Application of Economic Viability Limits (Art. 61-B)

After applying the buffers based on hydrographic geometries, the deficit areas associated with the marginal recomposition strips stipulated by the ‘Staircase Rule’ ($D_{\text{escadinha}}$) are calculated. The developed script acted as a legal framing filter and area limiter, executing the following logical stages. To safeguard the economic viability of small rural properties, legislation imposes a maximum cap on the commitment of the total area of the property (A_{total}) for the recomposition of consolidated areas. The model instituted a percentage limit variable (L_{trava}), conditioned on the size of the property in Fiscal Modules:

$$L_{\text{trava}} = 0.10 \times A_{\text{total}} \quad \text{if Fiscal Modules} \leq 2$$

$$L_{\text{trava}} = 0.20 \times A_{\text{total}} \quad \text{if } 2 < \text{Fiscal Modules} \leq 4$$

$$L_{\text{trava}} = \infty \quad (\text{no limit}), \text{ if Fiscal Modules} > 4$$

The requirement to restore past liability (R_{deficit}) was then defined by the lower value between the spatial deficit calculated in the “staircase” widths and the protective limit applicable to the property:

$$R_{\text{deficit}} = \min(D_{\text{escadinha}}, L_{\text{trava}})$$

3.5.2 Incorporation of Recent Deforestation and Full Restoration

Unlike the consolidated liability, native vegetation suppressions occurring after the July 22, 2008 milestone (APP_{UAS}) constitute a continuing infraction and do not enjoy partial amnesty or recomposition reducers. Consequently, the model imposed the full restoration of these areas. The final recomposition obligation (R_{APP}) was equated by the sum of the adjusted consolidated liability and the recent deforestation:

$$R_{APP} = R_{\text{deficit}} + APP_{UAS}$$

3.6 Consolidation of Mandatory In Loco Recomposition

A structural premise of the method consisted in the strict segregation between Permanent Preservation Area and Legal Reserve obligations, preventing undue overlap of areas destined for off-site environmental compensation (Environmental Reserve Credits). By legal determination, all preservation deficit and all Legal Reserve suppression occurring after 2008 must be obligatorily mitigated within the property perimeter itself. In this way, we synthesized the owner's material obligation through the Total In Loco Restoration variable (R_{total}), expressed by the linear summation of the physical recomposition requirements on the farm:

$$R_{\text{total}} = R_{APP} + R_{LR_insitu}$$

Where R_{LR_insitu} represents the portion of the Legal Reserve deficit generated by alternative land use after 2008, as modeled in the previous section. This consolidated variable provides the exact area metric that must be submitted to Degraded Area Recovery Plans.

3.7 Validation and Audit of Models

To ensure methodological integrity and fidelity to legal precepts, an audit protocol was established. The script was executed in successive batches containing hundreds of thousands of state

registrations. Processed results were subjected to a battery of matrix tests in Python language, which recreated independent logics aiming to detect inconsistencies (false-positives of exemption and overtaxation). This iterative debugging method ensured the identification and correction of any errors, validating the proposed algorithm for large-scale application in public policy.

3.8 Methodology for Cattle Supplier Classification

The classification of cattle suppliers was based on the integration of Animal Transit Guide (GTA) data with CAR, enabling the tracking of cattle movements across the supply chain. Properties are categorized into distinct tiers—Direct Suppliers or Indirect Suppliers—based on the predominant volume of their annual cattle transactions.

The analysis begins by cleaning the dataset to handle missing values and filtering out properties with no registered movement in a given year. For each property, we calculate the ratio of cattle sent to slaughterhouses (federal, state, or other inspection levels) relative to the total number of animals moved, ensuring an identification of the farm's primary economic role within the beef production cycle.

Under this methodological framework, a property is labeled a Direct Supplier if more than 50% of its total annual cattle volume is destined for slaughter. Conversely, properties that do not meet this threshold are classified as Indirect Suppliers and further subdivided into Tier 1 (T1) or Tier 2+ (T2+) based on their transaction profile. If over 50% of a property's movement involves Tier 1 transactions (direct sales to fattening farms), it is classified as T1; otherwise, it is assigned to the T2+ group, which represents deeper layers of the indirect supply chain. This tiered approach allows for a granular assessment of socio-environmental compliance, as it distinguishes between farms that finish cattle for slaughter and those primarily engaged in calving or rearing for other intermediaries.

4. Results

4.1 Rural Environmental Registry Status

As of March 2026, the CAR of Mato Grosso comprised 176,020 registered rural properties, covering approximately 82.3 million hectares.

Most of these registrations were classified as active (159,419 properties or 90.6% of area) while 16,601 registrations (9.4% of properties or 11.5% of area) were classified as inactive under the CANCELADO and INDEFERIDO statuses, indicating that they had been canceled or rejected following the analysis by the state environmental agency (SIMCAR, 2026). This usually occurs when properties have incomplete information, do not address the issues detected by the environmental agency within the given deadline (generally 90 days), or have an unjustified overlap with another property or identification of non-existence based on documentation or judicial determinations.

Despite ongoing efforts to validate the CARs, the system remains largely composed of self-declared information. More than half of all registrations (53.1%) are still awaiting validation, representing approximately 43.67 million hectares without official validation of their boundaries and land uses.

Table 2: Status of Rural Environmental Registrations in Mato Grosso, March 2026.

Registration Status	Number of Properties	% of Properties	Area (Mha)	% of Area
Active	159,419	90.6%	72.8	88.5%
Inactive (Canceled / Rejected)	16,601	9.4%	9.5	11.5%
Total	176,020	100.0%	82.3	100.0%

Validation of registrations to date has favored large properties. Currently, fully validated properties that have completed all stages of state verification total 42,094 registrations (26% of the active registrations) but cover 35.4% of all registered areas. This concentration of validated

areas relative to the total number of properties suggests that the validation process has so far advanced more quickly for large landholdings.

Small properties still make up the majority of pending statuses in part because approximately half of the small properties consist of plots located within agrarian reform settlements. Settlements require a distinct validation process, which considers the total area of the settlement prior to its subdivision into settlement plots (Article 12, Paragraph 1, of the FC). This process requires the involvement of the agencies responsible for the settlements—which, in the state, are INCRA or INTERMAT—because they possess the information regarding the exact division of the lots, the beneficiaries, and whether the LR will be established collectively for the entire settlement or maintained individually on each lot. In Mato Grosso, the SIMCAR-Assentamentos system was developed specifically for this process, becoming operational in late 2025 (State Decree 1,756/2025).

Within the group of properties that have already undergone validation, the reality of environmental compliance with FC emerges more clearly. We observed that 16.9% of all properties have already been notified of their environmental liabilities, but have not yet formally begun the recovery process. Only a very small percentage of owners, just over 3,800 properties, have completed the entire bureaucratic process and are currently in an active environmental regularization regime. The predominance of pending statuses, for 66.7% of the properties, means that environmental governance is held in a state of uncertainty, as landowners are unable to address their non-compliances until they have confirmation of them from the state.

In practical terms, the scenario in the first quarter of 2026 indicates a step in the right direction. Most registrations have already undergone an initial analysis or are being processed, with 45.9% of registrations awaiting initial verification. This is a result of the automated analysis and validation process that accelerated the process, promoted by the CAR Digital technology (Lopes, 2025).

Table 3: Active Registrations Breakdown by Status.

Registration Status	Properties	%	Area (Mha)	%
Pending Statuses (Subtotal)	117,325	66.7%	43.67	53.1%
AGUARDANDO_ANALISE	80,740	45.9%	22.97	27.9%
AGUARDANDO_COMPLEMENTACAO	27,464	15.6%	13.50	16.5%
EM_ANALISE	9,121	5.2%	7.13	8.7%
Validated Statuses (Subtotal)	42,094	23.9%	29.14	35.4%
AGUARDANDO_ENVIO_PRA	29,683	16.9%	16.79	20.4%
CAR_VALIDADO	8,201	4.7%	5.24	6.4%
CAR_VALIDADO_EM_REGULARIZACAO	3,824	2.2%	6.59	8.0%
AGUARDANDO_ANALISE_PRA	386	0.2%	0.53	0.7%
Active Registry (Total)	159,419	90.6%	72.80	88.5%

4.2 Rural Environmental Registry Categories

Among the active registrations, the majority were awaiting initial or subsequent analysis by the state agency. The category AGUARDANDO_ANALISE, which are the registrations that have not even provisionally evaluated, are the single largest category by property count, accounting for 80,740 properties (45.9% of all properties) and 22.97 million hectares (Mha) (27.9% of total area). Its proportionally lower area share relative to its property share was consistent with a concentration of smaller-sized holdings within this status. EM_ANALISE refers to registrations that have had their analysis initiated but not finalized, corresponding to 9,121 records (5.2% of properties) and an area of 7,13 Mha (8.7%). AGUARDANDO_COMPLEMENTACAO represents registrations that have already been analyzed, and the agency has identified pending issues, which have not yet been addressed by the responsible party, and accounted for 15.6% (27,464 properties) and 13,5 million hectares (16,5%). Taken together, these three pending statuses encompass 117,325 properties of registered properties (66,7%) and 43.67 Mha of total registered area (53,1%).

Properties with completed validation and approved analysis without deficits, CAR_VALIDADO, represent 8.201 properties (4.7%) or 5.24 Mha of registered area (6.4%). Those validated, approved and in active environmental regularization (CAR_VALIDADO_EM_REGULARIZACAO) added a further 3,824 properties (2.2%) and 6.59 Mha of area (8.0%). AGUARDANDO_ENVIO_PRA, assigned to properties that have completed CAR validation but for which the property owner has not

yet submitted their Environmental Regularization Program, represented 29,683 properties (16.9%) and 16,79 Mha of area (20.4% Mha). AGUARDANDO_ANALISE_PRA are properties that have been validated and have submitted the recovery project, which has not yet been approved by the environmental agency, with 386 properties (0.2%) amounting to an area of 532 thousand hectares (0.7%) in this category.

The categories CAR_VALIDADO_EM_REGULARIZACAO, AGUARDANDO_ENVIO_PRA, and AGUARDANDO_ANALISE_PRA include all validated properties that have LR or APP deficits. After the validation process, they are directed to the Environmental Regularization Program (PRA), so that the property representative must sign the Commitment Agreement that will address the deficit and submit the project for the recovery of degraded areas. These categories appear differentiated in the CAR to indicate the stage they are in, with CAR_VALIDADO_EM_REGULARIZACAO indicating those that have completed this procedure and supposedly started the recovery (Mato Grosso, 2017).

4.3 Structure of Properties by Size and Special Areas

The structure of properties by size is marked by concentration: large properties occupy two-thirds of the territory, while smallholdings hold a minimal share of the total area, although they constitute the largest number of properties—more than 6 times that of large properties.

To recap from Chapter 2, the classification of properties in Brazil as small, medium, or large follows laws from 1979 and 1993 and is defined by "Fiscal Modules" (FM), which vary between 30 and 100 hectares, depending on the municipality. For this work, we have included the concept of "Minifundio" for properties up to 1 fiscal module, as the Forest Code applies specific rules for these types of properties.

Table 4: Characterization of properties by size class (Fiscal Modules).

Size Class	N	% N	Area (ha)	% Area
Large (> 15)	14,827	8.4%	54,648,349	66.4%
Medium (4-15)	22,084	12.6%	15,387,756	18.7%
Small (1-4)	48,181	27.4%	8,668,590	10.5%
Minifundio (≤ 1)	90,928	51.7%	3,618,850	4.4%
Total	176,020	100.0%	82,323,544	100.0%

Table 5: Registry status by size class (Fiscal Modules).

Size Class	Pending	Validated	Inactive	Total
Large (> 15)	7,561	5,595	1,671	14,827
Medium (4 to 15)	11,434	8,515	2,135	22,084
Small (1 to 4)	29,812	13,011	5,358	48,181
Minifundio (≤ 1)	68,518	14,973	7,437	90,928
Total	117,325	42,094	16,601	176,020

In the context of compliance with FC, this study groups Conservation Units, Rural Settlements, and Quilombola Territories under the umbrella of "special areas". These areas are subject to differentiated legal regimes, under the FC and Law No. 9,985/2000, which regulates the National System of Nature Conservation Units (SNUC)—and establishes specific land-use restrictions, collective LR accounting mechanisms, and tailored management plans that set them apart from standard private properties. However, due to data limitations that prevent the automated assessment of these specific legal requirements, we apply the standard FC rules to these areas. This approach is consistent with prior studies and reflects common practice in large-scale analyses, where detailed information on differentiated legal regimes is often unavailable (Rajão et al., 2020; Sonda et al., 2023; Chiavari et al., 2024). It is important to clarify that the Brazilian model of conservation units allows, in some cases, for private properties to continue being used until the conservation unit "law," called the Management Plan, is created and effective expropriation occurs (when necessary) (Brazil, 2000). For this reason, there are private properties in this type of special area in Brazil.

Within these special areas, we identified 50,156 private land holdings covering approximately 6.46 million hectares. Rural settlements account for the overwhelming majority of the registries, making up 91.2% of the total, yet they occupy roughly 47.8% of the mapped area, totaling nearly 3.09 million hectares. This dynamic is heavily driven by smallholdings, as minifundios alone represent 73.4% of all properties in these special areas. On the flip side, Conservation Units present the exact opposite pattern: they make up merely 8.8% of the registries but control most of the area in special areas, encompassing nearly 3.38 million hectares, which accounts for 52.2% of the total area. This happens because conservation units

can contain private properties (Brazil, 2000), which will naturally vary in size. The data also notes the presence of Quilombola Territories, though they represent a statistically negligible footprint in this specific registry subset, with only two smallholding properties recorded.

Table 6: Characterization of rural properties in special areas.

Category	Size Class	N	% N	Area (ha)	% Area
Conservation Units	LARGE (>15)	563	1.1%	2,591,093	40.1%
	MEDIUM (4-15)	825	1.6%	537,573	8.3%
	SMALL (1-4)	1,259	2.5%	210,260	3.2%
	MINIFUNDIO (≤ 1)	1,742	3.5%	39,636	0.6%
<i>Subtotal</i>		<i>4,389</i>	<i>8.8%</i>	<i>3,378,563</i>	<i>52.2%</i>
Quilombola Territories	MINIFUNDIO (≤ 1)	2	0.0%	2	0.0%
<i>Subtotal</i>		<i>2</i>	<i>0.0%</i>	<i>2</i>	<i>0.0%</i>
Rural Settlements	LARGE (>15)	72	0.1%	158,722	2.4%
	MEDIUM (4-15)	338	0.7%	220,024	3.4%
	SMALL (1-4)	8,534	17.0%	1,081,178	16.7%
	MINIFUNDIO (≤ 1)	36,821	73.4%	1,629,123	25.2%
<i>Subtotal</i>		<i>45,765</i>	<i>91.2%</i>	<i>3,089,047</i>	<i>47.8%</i>
Total		50,156	100.0%	6,467,612	100.0%

4.4 Legal Reserve: Deficits, Surpluses, and Balance

To calculate whether there is a deficit, surplus, or compliance with the Forest Code standards regarding the LR, several steps are necessary. The general rule for Mato Grosso, requiring 80% conservation for the Forest typology and 35% for the Cerrado, must be combined with the exemptions of articles 12, 67, and 68 of the Code, which can reduce this percentage (Brazil, 2012).

The calculations for LR also depend on the area of the property, contained within the limits of the CAR, as well as the existing native vegetation. The total area of primary and regenerating vegetation that falls into this native vegetation is 59,113,117 hectares, with 49,868,334 hectares of primary vegetation and 9,244,783 hectares of secondary vegetation in Mato Grosso.

Our analysis of LR obligations reveals how the rules of the FC adapt to the history of land occupation in Mato Grosso, providing exemptions that are meant to account for the diversity of producers' trajectories. We estimate that the maximum FC deficit, without the exemptions that account for this history, would be 9.98 million hectares across 74,823 properties.

The most important of these exemptions, detailed in Article 67, seeks to reduce the burden of

reduced production areas on smallholders. Article 67 results in the exemption of more than 54,000 small producers, —those with properties up to 4 fiscal modules— from the obligation to restore LR areas deforested before 2008. This provision reduces the state’s LR deficit by 1.41 million hectares. Deforestation after 2008, even on small properties, must follow the basic rules for LR.

Similarly, Article 68 exemptions —which recognize compliance with the legal requirements at the time the land was converted— eliminates the need for restoration of an additional 2.27 million hectares for properties that maintained the minimum required LR prior to May 2000. This mechanism accommodates changes in legal requirements through time. When combined with further adjustments provided under Article 12 which cut the deficit by 1.79 million hectares—applicable to municipalities where more than 50% of the territory consists of protected areas, such as Indigenous lands and public Conservation Units (Alto Boa Vista, Apiacás, Comodoro, Juína, Novo Santo Antônio, and Tangará da Serra)— the effective LR deficit is reduced to 4.51 million hectares, distributed across 43,571 properties.

The adjusted distribution of environmental liabilities tells a story of concentrated responsibility. Legal exemptions have practically eliminated the LR deficits for small farmers, shifting nearly the entire restoration obligation (88.6%) to large landowners. Consequently, the challenge of forest conservation in Mato Grosso is no longer a highly dispersed issue, but rather a task concentrated in the hands of a smaller group of properties, but one that is territorially dominant, as table 7 shows.

Table 7: LR deficit per size class (mean, median and total) in Mato Grosso.

Size Class	Properties (n)	Instrument	Mean deficit (ha)	Median deficit (ha)	Total deficit (ha)
Minifundio (≤ 1)	18,195	LR	8.70	4.30	158,296
Small (1-4)	8,906	LR	23.80	9.30	211,963
Medium (4-15)	10,940	LR	136.40	97.70	1,492,216
Large (> 15)	5,530	LR	479.80	294.50	2,653,294

The results show that while a significant portion of the state properties preserve more than required, about a quarter of them need to be addressed LR deficits to come into compliance with the FC, whether through planting or the use of financial compensation mechanisms.

4.5 Pathways to Legal Reserve Compliance: Restoration versus Compensation

The FC provides for two ways to address LR deficits — on-property restoration and off-property compensation, which are available to the property owner depending on the timing of the deforestation that generated the deficit. From the net LR liability, we identified that 1.33 million hectares distributed across 19,304 properties must be restored within property boundaries as these areas were deforested after the 2008 legal cutoff established by the FC. In contrast, the remaining 3.18 million hectares, associated with 40,223 properties, resulting from land conversion prior to 2008, may be regularized through compensation mechanisms without the need for active ecological restoration. These mechanisms include acquisition of the CRA, the establishment of environmental easement, or purchase or donation of land within a Conservation Units pending regularization (Brazil, 2012).

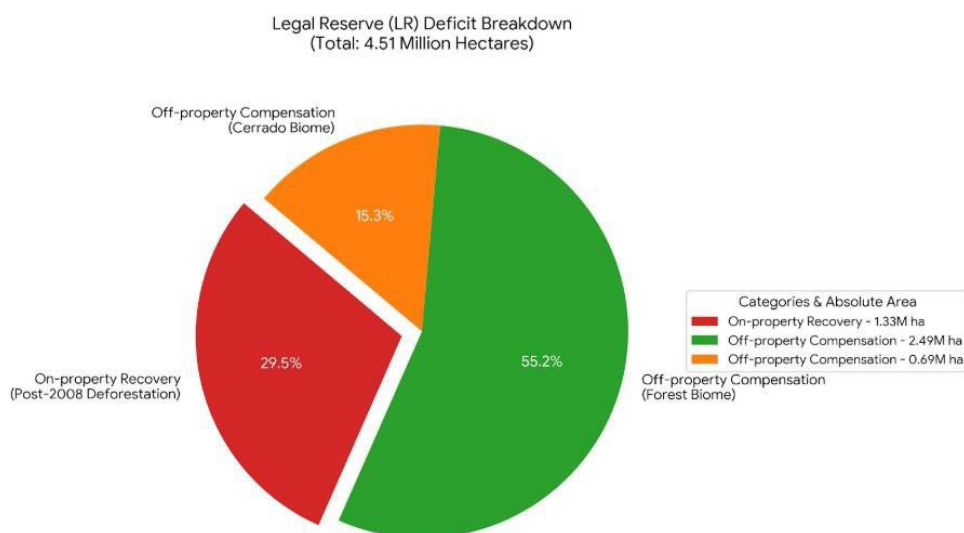


Figure 4: Legal reserve deficit breakdown in Mato Grosso.

For rural landowners, the choice of compliance path depends directly on the nature of their liabilities. Almost half of the properties with deficits can resolve their situation entirely through compensation, without giving up productive areas. Another 18% must restore vegetation within the property. A significant portion of producers (15,956), however, face a mixed situation, requiring both financial planning for the purchase of compensation quotas and operational effort

for planting.

The distribution of these deficits reinforces the thesis that environmental regularization is a matter of scale. Only 600 properties (0.34% of the state's total properties) concentrate almost 25% of the state's entire LR deficit (4.51 Mha), with obligations exceeding 1,000 hectares per property. This data suggests that the success of forest governance depends less on massive actions and more on strategies focused on this group of large landowners, who have the power to significantly transform the state's conservation indicators.

The forestry asset market in Mato Grosso, which is the area that is eligible to be used for compensation, shows substantial potential. The state's total LR surplus amounts to approximately 13.8 million hectares distributed across 19,304 properties. In terms of vegetation types, the surplus in the Forest is 5.65 million hectares, and in the Cerrado, it is 7.84 million hectares.

Of the total LR deficit, 3.18 million hectares originate from consolidated deforestation on areas that are eligible for regularization through off-property compensation mechanisms. This includes 2.49 million hectares in forest and 0.69 million hectares for Cerrado. The compensation market shows a favorable outlook in terms of potential availability of areas that can serve for compensation through Environmental Reserve Quotas (CRA) or environmental easement amounts, given the 13.8 million hectares of native vegetation surplus available across the state.

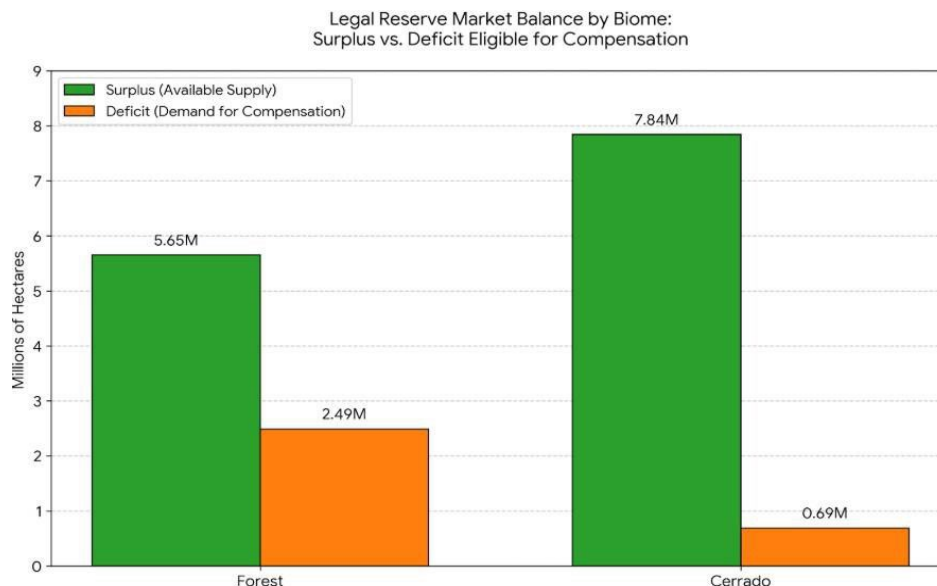


Figure 5: Deficit of legal reserve and supply vs. demand eligible for compensation by biome in Mato Grosso.

The distribution of land eligible for the CRA demonstrates a favorable market scenario. With a net surplus of more than 10.62 million hectares relative to demand, the available supply of native vegetation is sufficient to meet the state’s need for compensation through CRA transactions or environmental easements, though access to these areas for use as CRA will be dependent on market factors and operationalization of environmental agency systems for compensation.

The surplus estimate incorporates the provisions of Article 68, paragraph 2, of the FC. Under this rule, properties located in the Legal Amazon that maintain more more than 50% of native vegetation in forest and 20% in cerrado — and did not exceed the legal deforestation limits in force at the time of conversion (in this case, prior to 2000) are permitted to allocate their surplus LR area towards compensation mechanisms including the CRA and environmental easement, thereby expanding the effective supply of eligible areas for market-based compliance (Brazil, 2012).

Below is a table showing the LR deficit and surplus by property size class.

Table 8: Legal Reserve obligations and compliance in Mato Grosso

Size Class	Required (ha)	Gross Deficit	Adj. Deficit	To Restore	To Compensate	Surplus (ha)
LARGE (>15)	29,466,758	7,371,251	2,653,051	639,435	2,013,616	9,224,772
MEDIUM (4-15)	8,005,995	1,093,821	1,492,575	338,409	1,154,166	1,959,121
MINIFUNDIO (≤1)	742,311	571,752	157,574	152,645	4,929	855,033
SMALL (1-4)	2,294,553	941,245	211,858	202,966	8,892	1,793,241
Total	40,509,617	9,978,070	4,515,058	1,333,455	3,181,603	13,832,168

4.6 Permanent Preservation Area Obligations

Permanent Preservation Areas, as defined under Articles 4 of the Forest Code, totaled 4.55 million hectares across all registered properties in Mato Grosso. 2.88 Mha of the required area were under native or secondary vegetation, leaving a gross deficit of 1.67 million hectares among active registrations. After applying the Forest Code exemptions (article 61-A), known as the “staircase rule”, the net deficit, which requires effective on-site restoration, is 1.13 million hectares distributed across 128,524 properties (73% of all properties).

The deficit distribution across property size classes showed distinct patterns. Although the APP deficit in minifundios is only about 46,600 hectares in absolute terms, the non-compliance rate in these properties is extremely high (79.2%). In contrast, large properties account for 715,873 hectares of the net APP deficit (63.3% of the state's total APP restoration liability).

Table 9: APP area of requirements, gross and net deficit per size class.

Size Class	APP Gross Deficit (ha)	APP Net Deficit (ha)	Mean Net deficit (ha)	Properties Net Deficit (n)
Large (> 15 FM)	2,883,806	715,873	56.60	13,281
Medium (4-15 FM)	858,649	233,701	14.30	18,125
Small (1-4 FM)	547,441	133,924	6.60	37,475
Minifundio ($\leq$ 1 FM)	263,399	46,653	2.20	59,773
Total	4,553,295	1,130,151	79.7	128,524

4.7 Combined Compliance

Cross-referencing LR and APP compliance status showed that only 35,971 properties (20.4%), covering 7.44 Mha achieved simultaneous full compliance with both obligations. The dominant configuration was LR-compliant with APP deficit, affecting 132,449 properties (75.2%) and 54.9 Mha. Properties non-compliant on both dimensions (LR and APP) numbered 3,542 yet collectively covered 1.78 Mha.

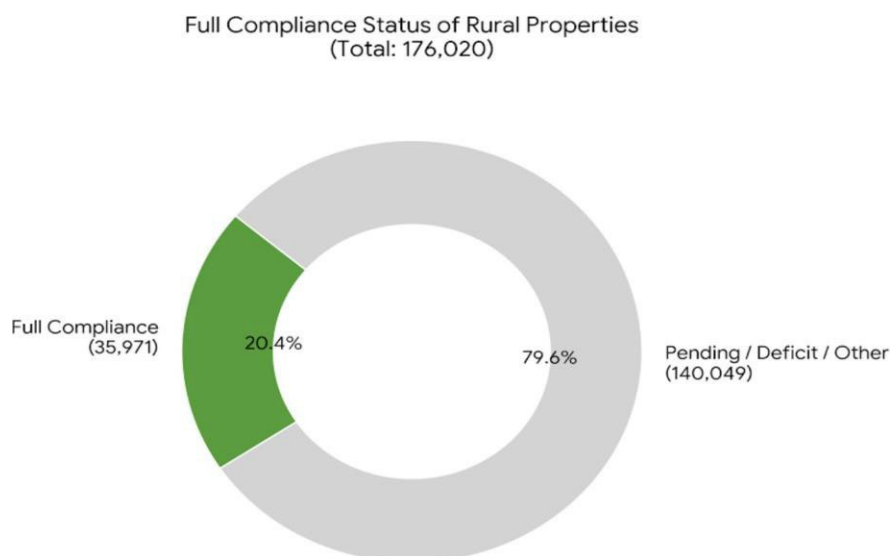


Figure 6: Full compliance status of rural properties in Mato Grosso.

However, the total deficit represents only 6.85% of the total area of private properties registered in the state (82.32 million hectares).

This scenario demonstrates that the APP is the main bottleneck for sustainability in Mato Grosso. While LR can be resolved in many cases with financial transactions, the recovery of 1.13 million hectares of APP requires a significant forest restoration effort, as it affects most properties (132,449). This obligation demands not only political will, but also an infrastructure for seedling production and technical assistance that reaches all productive strata. The data below shows the aggregate deficit of RL and APP by property class, reinforcing this statement.

Table 10: Combined Forest Code liabilities by size class.

Size Class	Gross Deficit (ha)	Adj. Deficit (ha)	To Restore (ha)	To Compensate (ha)
LARGE (>15)	8,210,127	3,368,924	1,355,308	2,013,616
MEDIUM (4-15)	1,410,464	1,726,276	572,110	1,154,166
SMALL (1-4)	1,260,954	345,782	336,890	8,892
MINIFUNDIO (≤ 1)	769,882	204,228	199,298	4,929
Total	11,651,427	5,645,210	2,463,606	3,181,603

The chart below (Figure 7) summarizes the number of areas by type of compliance required by the Forest Code for the identified deficits.

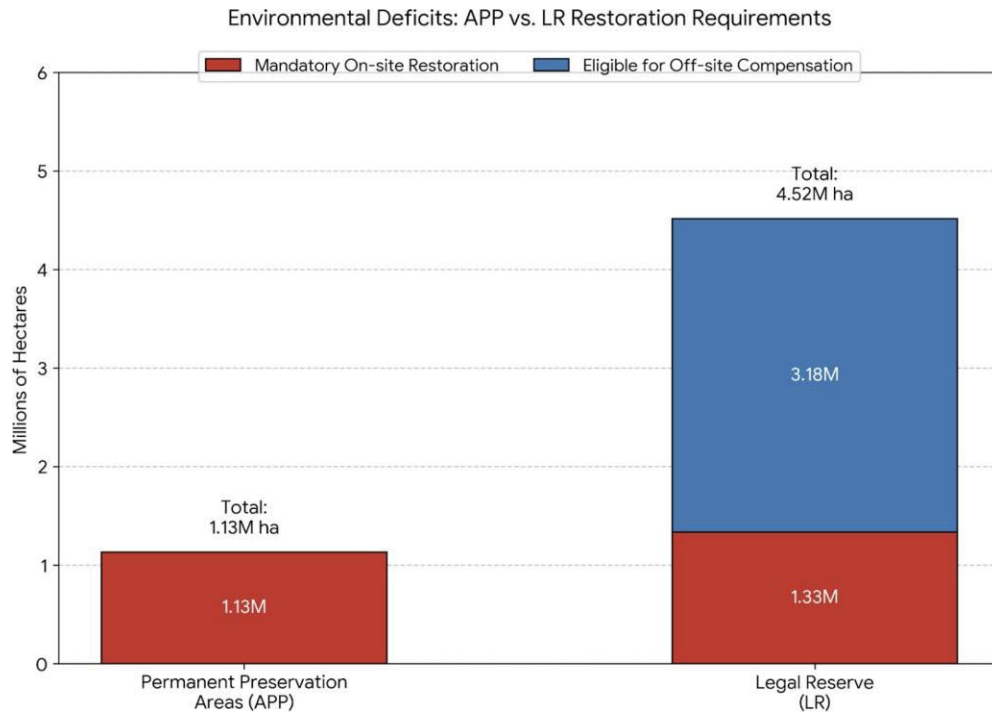


Figure 7: Liability for Restoration and Compensation.

The average combined deficit per property class demonstrates a concentration of liabilities in large properties.

Table 11: Mean total deficit (APP+RL) per size class.

Size Class	Total Mean Deficit (ha)
LARGE (>15)	235.50
MEDIUM (4-15)	81.90
SMALL (1-4)	11.00
MINIFUNDIO (≤1)	3.90

When examining the combined Forest Code liabilities by size in special areas (conservation units, settlements, and quilombola), encompassing both LR and APP, the data reveal how legal frameworks resize the actual environmental burden. While these special areas carry a massive total gross deficit of 1,222,558 hectares, applying legal adjustments and exemptions drastically reduces this to an adjusted deficit of just 256,216 hectares. However, the composition of this

remaining liability is particularly challenging from a public policy perspective. Out of the adjusted total, the vast majority—215,708 hectares—must be physically restored in-situ, leaving only 40,508 hectares eligible for financial compensation. This physical restoration burden falls disproportionately on agrarian reform communities, with rural settlements alone needing to directly restore 173,157 hectares. This reality underscores technical and financial hurdles these socially vulnerable populations face in achieving full environmental compliance compared to standard private landowners.

Table 12: Combined Forest Code liabilities (LR + APP) by special areas.

Category	Size Class	Gross Deficit	Adj. Deficit	To Restore	To Compensate
Conservation Units	LARGE (>15)	570,046	33,195	23,252	9,943
	MEDIUM (4-15)	20,746	20,304	14,304	5,999
	MINIFUNDIO (≤1)	3,087	1,017	912	105
	SMALL (1-4)	11,035	4,189	4,082	107
<i>Subtotal</i>		<i>604,915</i>	<i>58,705</i>	<i>42,551</i>	<i>16,154</i>
Quilombola	MINIFUNDIO (≤1)	0	0	0	0
<i>Subtotal</i>		<i>0</i>	<i>0</i>	<i>0</i>	<i>0</i>
Rural Settlements	LARGE (>15)	13,614	15,696	5,069	10,627
	MEDIUM (4-15)	21,803	20,927	11,083	9,844
	MINIFUNDIO (≤1)	408,004	106,100	103,863	2,237
	SMALL (1-4)	174,222	54,788	53,142	1,646
<i>Subtotal</i>		<i>617,643</i>	<i>197,511</i>	<i>173,157</i>	<i>24,354</i>
Total		1,222,558	256,216	215,708	40,508

Comparing the data with your geographic region, the following 20 municipalities have the largest aggregate total deficits (Table 13). Detailed information on the deficits in all municipalities of Mato Grosso can be found in the supplementary material.

Table 13: Top 20 municipalities with the largest aggregate deficit.

Municipality	Properties	LR Surplus	LR Deficit	APP Deficit	Total Deficit
Nova Ubiratã	2,373	140,434	171,626	7,854	179,481
Sorriso	3,189	36,340	166,062	12,744	178,807
Juara	2,589	215,870	88,299	63,257	151,557
Paranatinga	2,172	427,514	88,070	51,068	139,138
Querência	2,355	92,241	121,433	9,843	131,276

Municipality	Properties	LR Surplus	LR Deficit	APP Deficit	Total Deficit
Brasnorte	2,486	182,750	108,395	19,032	127,427
Nova Maringá	1,343	131,888	115,188	8,955	124,144
Gaúcha Do Norte	1,600	111,139	106,677	10,553	117,231
Colniza	3,017	229,256	80,155	36,086	116,241
Nova Mutum	2,169	56,968	104,324	9,626	113,950
Alta Floresta	3,138	76,593	63,683	49,472	113,155
Porto Dos Gaúchos	1,222	54,010	100,683	11,606	112,290
Diamantino	1,831	65,371	106,069	4,460	110,529
São Félix Do Araguaia	1,597	348,921	90,329	17,975	108,305
Nova Bandeirantes	2,401	75,907	60,875	34,892	95,768
Peixoto De Azevedo	2,871	111,516	67,986	25,523	93,510
Ipiranga Do Norte	1,129	9,017	88,277	4,070	92,347
Vila Bela da S. Trindade	2,467	302,213	66,798	24,948	91,746
Vila Rica	2,460	122,618	54,713	36,296	91,010
Tabaporã	1,907	73,404	77,571	11,477	89,048

After presenting the results of the properties' compliance with the rules of the Forest Code, we will move on to the impact of this compliance on the cattle supply chain.

4.8 Cattle Supply Chain Analysis

To verify which properties are part of the beef supply chain, we cross-referenced the information from the Mato Grosso Animal Transit Guide between 2021 and 2024 with the registry database. We identified 47,015 properties linked to the sector, but only a small fraction (7.4%) sold animals directly to slaughterhouses frequently. Most properties sell to other farms. For this study, we classified them as indirect suppliers.

Table 14: Characterization of rural properties in the MT cattle supply chain.

Category	Size Class	N	% N	Area (ha)	% Area
Direct Supplier	LARGE (>15)	598	1.3%	2,257,554	12.6%

Category	Size Class	N	% N	Area (ha)	% Area
	MEDIUM (4-15)	655	1.4%	449,125	2.5%
	SMALL (1-4)	971	2.1%	180,164	1.0%
	MINIFUNDIO (≤ 1)	1,280	2.7%	53,032	0.3%
<i>Subtotal</i>		<i>3,504</i>	<i>7.4%</i>	<i>2,939,875</i>	<i>16.4%</i>
Indirect Supplier - T1	LARGE (>15)	610	1.3%	2,219,370	12.4%
	MEDIUM (4-15)	866	1.8%	620,980	3.5%
	SMALL (1-4)	1,835	3.9%	332,749	1.9%
	MINIFUNDIO (≤ 1)	3,636	7.7%	150,145	0.8%
<i>Subtotal</i>		<i>6,947</i>	<i>14.8%</i>	<i>3,323,244</i>	<i>18.6%</i>
Indirect Supplier - T2+	LARGE (>15)	1,950	4.2%	6,924,790	38.7%
	MEDIUM (4-15)	3,283	7.0%	2,192,741	12.2%
	SMALL (1-4)	9,265	19.7%	1,581,274	8.8%
	MINIFUNDIO (≤ 1)	22,066	46.9%	934,260	5.2%
<i>Subtotal</i>		<i>36,564</i>	<i>77.8%</i>	<i>11,633,065</i>	<i>65.0%</i>
Total		47,015	100.0%	17,896,183	100.0%

The total environmental liability mapped in this supply chain requiring on-site restoration reaches 675,362 hectares, but its distribution reveals a strategic flaw in current controls. Direct suppliers, who are the only ones monitored by current market agreements, account for less than 15% of this liability. Most of the environmental deficit to restore is "hidden" within T1 and T2+ indirect suppliers, who concentrate 85,24% of the restoration obligations. This demonstrates that current monitoring protocols are focused on the final link in a chain where animals are born and raised in areas with significant environmental liabilities.

Table 15: Combined Forest Code liabilities (LR+APP) in cattle chain.

Category	Size Class	Gross Deficit	Adj. Deficit	To Restore	To Compensate
Direct Supplier	LARGE (>15 MF)	336,718	232,999	75,219	157,781
	MEDIUM (4–15 MF)	50,157	75,223	17,608	57,615
	SMALL (1–4 MF)	27,368	5,768	5,574	194
	MINIFUNDIO (≤ 1 MF)	9,818	1,347	1,275	72
<i>Subtotal</i>		<i>424,060</i>	<i>315,337</i>	<i>99,676</i>	<i>215,662</i>
Indirect - T1	LARGE (>15 MF)	313,575	175,563	80,897	94,666
	MEDIUM (4–15 MF)	68,125	78,036	29,904	48,132
	SMALL (1–4 MF)	51,940	13,876	13,678	198
	MINIFUNDIO (≤ 1 MF)	32,066	7,595	7,471	124
<i>Subtotal</i>		<i>465,706</i>	<i>275,069</i>	<i>131,949</i>	<i>143,121</i>

Category	Size Class	Gross Deficit	Adj. Deficit	To Restore	To Compensate
Indirect - T2+	LARGE (>15 MF)	1,037,265	444,078	200,140	243,938
	MEDIUM (4–15 MF)	227,833	263,782	110,100	153,681
	SMALL (1–4 MF)	267,356	75,837	73,824	2,014
	MINIFUNDIO (≤1 MF)	234,414	60,681	59,674	1,007
<i>Subtotal</i>		<i>1,766,867</i>	<i>844,378</i>	<i>443,738</i>	<i>400,640</i>
Total		2,656,633	1,434,784	675,362	759,422

This indirect supply structure has a strong social base, where almost half of the properties are smallholdings (minifundios). However, the volume of total deficit is not held by them. Although numerous, small producers hold only a fraction of the total liability (4,85%), whereas large properties operating as indirect suppliers concentrate almost 60% of the deficit. This imbalance shows that regularizing cattle ranching requires solutions that differentiate social support for small producers from rigorous restoration requirements for the chain's large links.

Ultimately, the data confirms that environmental control of cattle ranching in Mato Grosso will only be effective when traceability reaches the true origin of the animals. The reliance on indirect suppliers creates an "environmental leakage" that compromises the sector's reputation. Integrating the CAR with cattle movement data is not just an enforcement measure, but a market necessity to ensure that the environmental liability of more than 600,000 hectares is addressed at all levels of production, and not just at the time of slaughter.

5. Discussion

This thesis provides a novel analysis of the outcomes of implementing environmental legislation on private properties and explores the impacts for the cattle production chain in the state of Mato Grosso. By cross-referencing state-scale active cartographic data from the Rural Environmental Registry (CAR) with the protection and exemption mandates established by the Forest Code, our findings highlight the complexity of reconciling the region's significant agricultural expansion with the necessity of conserving and restoring native vegetation. Here, we discuss our main findings by interpreting the tensions between in-situ conservation and financial compensation mechanisms, while also evaluating how monitoring gaps regarding indirect cattle suppliers compromise both ecological integrity and the reputation of Mato Grosso's agribusiness.

These findings must be interpreted in light of the current status of the CAR system in Mato Grosso, which remains in a transitional phase between initial registration and full regulatory enforcement. While adherence to the registry is widespread, a substantial share of properties (66.7%) is still awaiting validation by the state environmental agency. This gap between self-declaration and official verification limits the immediate effectiveness of CAR as a governance instrument, as many landholders have not yet received formal confirmation of their environmental status.

Our results show the validation process has also progressed unevenly across property types, advancing more quickly among larger holdings while smaller properties—particularly those linked to agrarian reform settlements—face more complex administrative requirements. This uneven pace contributes to a broader condition of regulatory uncertainty, in which many producers are aware of potential liabilities but are not yet fully integrated into formal regularization pathways.

Against this backdrop, the geospatial analysis of the CAR data in the state of Mato Grosso reveals a mix of surpluses and deficits of LR and APP on farms across the state, creating several key strategic opportunities for public management. Overall we show that environmental regularization, through compliance with the Forest Code rules, is a feasible task and may be less of a burden to rural producers than reported in other studies (Rajao et al., 2020), provided that all exceptions to the Forest Code are fully applied, which requires proper technical assistance to the landowner. In addition, our GIS analysis shows that irregular deforestation in MT is concentrated, suggesting that focused public policies can achieve significant results with greater operational efficiency.

However, the complexity in applying all of Forest Code rules is a major roadblock to achieving full compliance across the state. Even public agencies may struggle to fully interpret and operationalize all of its provisions. Previous modeling efforts in academic literature (Britaldo Soares-Filho et al., 2014; Rajao 2020), captured many core elements but necessarily relied on simplifications and omitted important legal nuances that we were able to capture here. In particular, key provisions related to Article 68—such as recognition of historical LR thresholds of 50% in forest areas and 20% for the Cerrado until May 2000 in the Legal Amazon— have often not been fully incorporated into analytical frameworks.

The model that we developed for this study is one of the most precise publicly available Forest Code model to date. The accuracy of our model provides the foundation for the development of public and private sector strategies to support increased compliance with the Forest Code. For example, our precisely indicates areas requiring recovery within each farm, considering exemptions that limit the need for restoration in consolidated areas across large, medium, and small properties, such as those provided for in Articles 12, 61-A, 67 and 68 of the FC. We also used a more precise hydrographic database than previous academic models, taking advantage of important advancements in availability of public datasets relevant for FC evaluation over the last several years, and which allowed for better application of the "staircase rule" on the recovery of APPs in consolidated areas, as contained in Article 61-A. By precisely assessing compliance gaps at the property level, we show spatial distributions patterns and characterize

different types and pervasiveness of non-compliance, which can help policymakers target interventions and resources.

Additionally, by cross-referencing cattle transit data, we found a deficit in LR post-2008, suggesting that direct and indirect suppliers have not yet fully adapted to the rules of the agreements. Property-level analysis conducted in this work allows the Federal Prosecutor's Office, the State Government, slaughterhouses, and ranchers' associations to develop more precise policies to support farmers who can utilize instruments for reintegration into the legal chain and in the process of complying with the FC, avoiding blockages due to sectoral agreements and the imposition of fines

5.1 Comparison with existing literature

A comparison of the results obtained in this study with the benchmark research conducted by Soares-Filho et al. (2014) and Rajão et al. (2020) reveals significant methodological advances, as well as important convergences and divergences regarding the diagnosis of environmental regularization in Mato Grosso. In terms of scale and unit of analysis, the pioneering study by Soares-Filho et al. (2014) was constrained by the absence of a unified national land tenure database at the time, which required the adoption of micro-watersheds as spatial proxies to estimate the application of the law. In contrast, both the present study and the work by Rajão et al. (2020) overcame this limitation by utilizing the exact vector boundaries of private properties in the Rural Environmental Registry (CAR). By employing property-level modeling with high-resolution spatial data (e.g., five-meter resolution matrices), these more recent approaches provide more accurate representation of land tenure structure and finer-grained assessment of compliance dynamics across the Amazon region.

Regarding the calculation of the Legal Reserve (LR), all three studies are methodologically aligned in basing protection requirements proportionally on the original phytophysiology (vegetation type) present on each property, distinguishing between the 80% requirement for forest areas and the 35% requirement for Cerrado areas. However, our work deepens this quantification by rigorously applying the amnesty rules set forth in Articles 67 and 68 of the Forest Code. This results in an adjusted net LR deficit of 4.51 million hectares, distributed across 43,571 properties,

and a Permanent Preservation Area (APP) liability reaching 1.13 million hectares that must be mandatorily restored in 128,524 properties. The application of these rules in our study also made it possible to accurately estimate the LR surplus, which was calculated at 13.8 million hectares.

While we focus on the state of MT, Soares-Filho estimated the impact of the Forest Code on a national scale, pointing out that the consolidation of the new legislation reduced the total area to be restored in Brazil from 50 Mha hectares to 21 Mha hectares (Soares-Filho et al., 2014). Of this national total, 78% (16,38 Mha), correspond to LR deficits, while the remaining 22% (4,62 Mha) relate to APP liabilities (Soares-Filho et al., 2014). When disaggregated to Mato Grosso, the same study projected that the projected Legal Reserve demand was set at approximately 2,900,000 hectares. Their estimate differs substantially from our estimate of 4,515,057 hectares. This difference of approximately 1.6 million hectares (55%) can be largely attributed to advances in data availability—particularly the use of property-level CAR boundaries and high-resolution spatial data—as well as the explicit incorporation of key legal provisions, such as Article 68, which were not fully captured in earlier models, and also to the deforestation that occurred over 10 years.

Although the primary scope of the 2020 study by Rajão and his team focused on the contamination of supply chains at the national level, the authors validated their spatial macro-models through intercomparison exercises designed for specific municipalities in Mato Grosso (Rajão et al., 2020). Analyzing our municipal database, we found that the areas required for conservation reach 1,361,049 hectares in the municipality of Juara, 562,683 hectares in the municipality of Juína, and 212,087 hectares in the municipality of Poxoréu. These values prove the accuracy of our property-level model and fall within the same order of territorial complexity as the model used in Rajão in 2020, which estimated the total sum of forest liabilities and surpluses in these same three municipalities at 2,150,000 hectares, 980,000 hectares, and 450,000 hectares, respectively (Rajão et al., 2020). However, there was a substantial difference, as our model applies the previous LR rules in accordance with Article 68, which reduces the requirement for several properties in municipalities with older agricultural and livestock development, while this was excluded from Rajão et al 2020

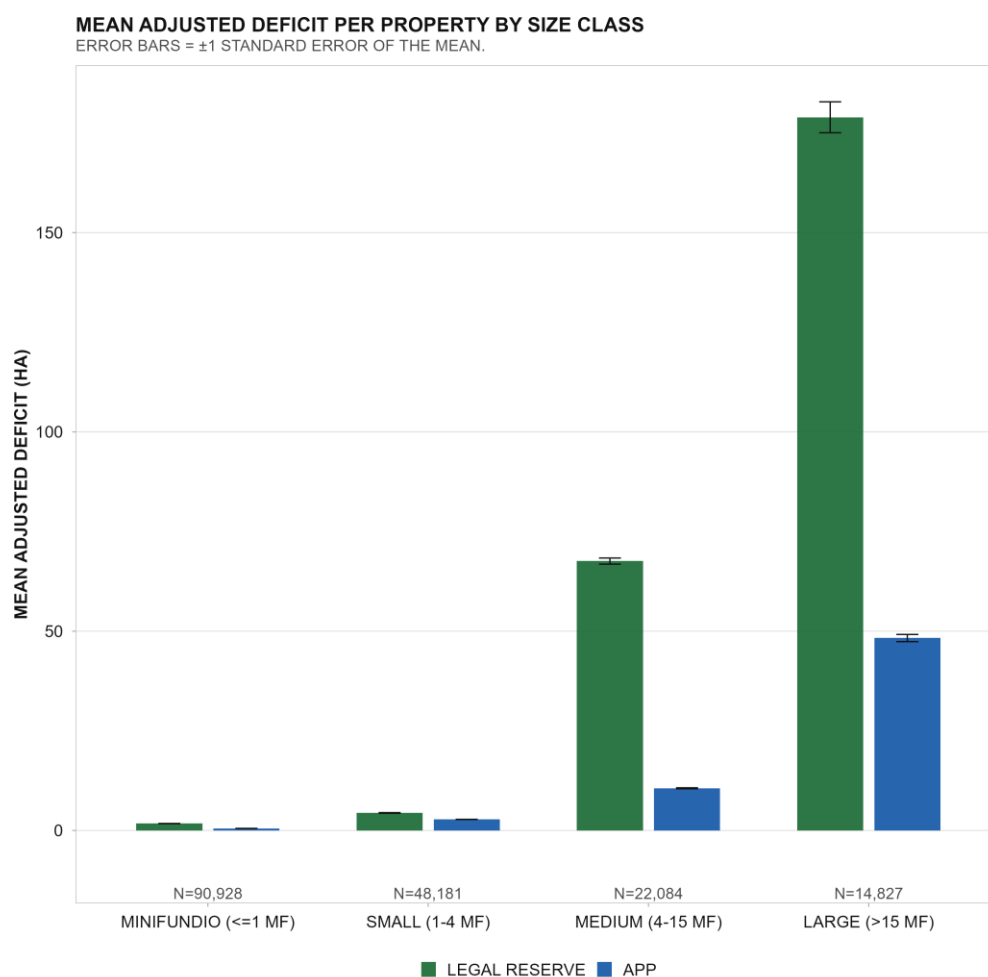
The refinement of the liability in our regional model corroborates Soares-Filho's thesis, which

demonstrated in its global model the drastic reduction of environmental liability promoted by the legislation. The national liability fell substantially following the Forest Code amnesties, disproportionately transferring the burden of restoring deforested areas to medium and large rural properties (Soares-Filho et al., 2014).

5.2 The strategy of concentration and efficiency through priority municipalities

We found the spatial distribution of environmental liabilities in Mato Grosso is highly uneven in relation to property size, creating a clear opportunity for targeted and cost-effective policy designs; rather than being broadly dispersed, environmental deficits are concentrated both geographically and across property types.

This concentration is closely tied to the structure of landholding in the state. Large properties, although few in number, dominate the agricultural landscape in terms of area, and therefore, hold the majority of the liabilities accumulating more than 58% of the Legal Reserve and more than 63% of the APP deficits. Smallholders on the other side, despite representing the majority of the properties, occupy a relatively small share of total area. The legal architecture of the Forest Code further amplifies this asymmetry: exemptions for pre-2008 deforestation and specific provisions for smallholders significantly reduce their formal liabilities, shifting the burden of compliance toward a smaller group of large landowners. In practice, this transforms environmental regularization into a process that hinges on engaging a relatively limited number of actors with disproportionate territorial influence.



SOURCE: SIMCAR/MT (MARCH 2026).

Figure 8: Mean deficit per property by size class.

At the same time, this concentration does not eliminate the need for differentiated policy approaches. While large properties account for most of the deficit, they are better positioned to use compensation mechanisms. Of the 2.65 million hectares of Legal Reserve deficit in this group, almost 2 million (74.6%) can be resolved through the acquisition of CRAs or other forms of compensation. This is a purely financial solution that does not require the return of productive land to nature (Rajão et al., 2020). Smallholders—particularly those in rural settlements and other special areas—face a different set of challenges. In these contexts, liabilities are smaller in absolute terms but are more likely to require mandatory on-site restoration, because excess deforestation prior to 2008, the period for which CRA is offered as a solution, was exempted by the provisions under Article 67. These properties are often operating under conditions of limited

technical and financial capacity, increasing the relative burden of remedying these noncompliances for these properties. As a result, these groups depend more heavily on state support, including rural extension services and targeted financing, to achieve compliance.

Together, these patterns suggest that an effective governance strategy must operate on two fronts. On one hand, enforcement and market-based mechanisms should be concentrated on large properties and priority municipalities, where the potential gains in compliance in terms of total area are greatest. On the other hand, social and technical support policies are essential for smaller and more vulnerable producers, for whom compliance is less a matter of financial optimization and more a question of feasibility.

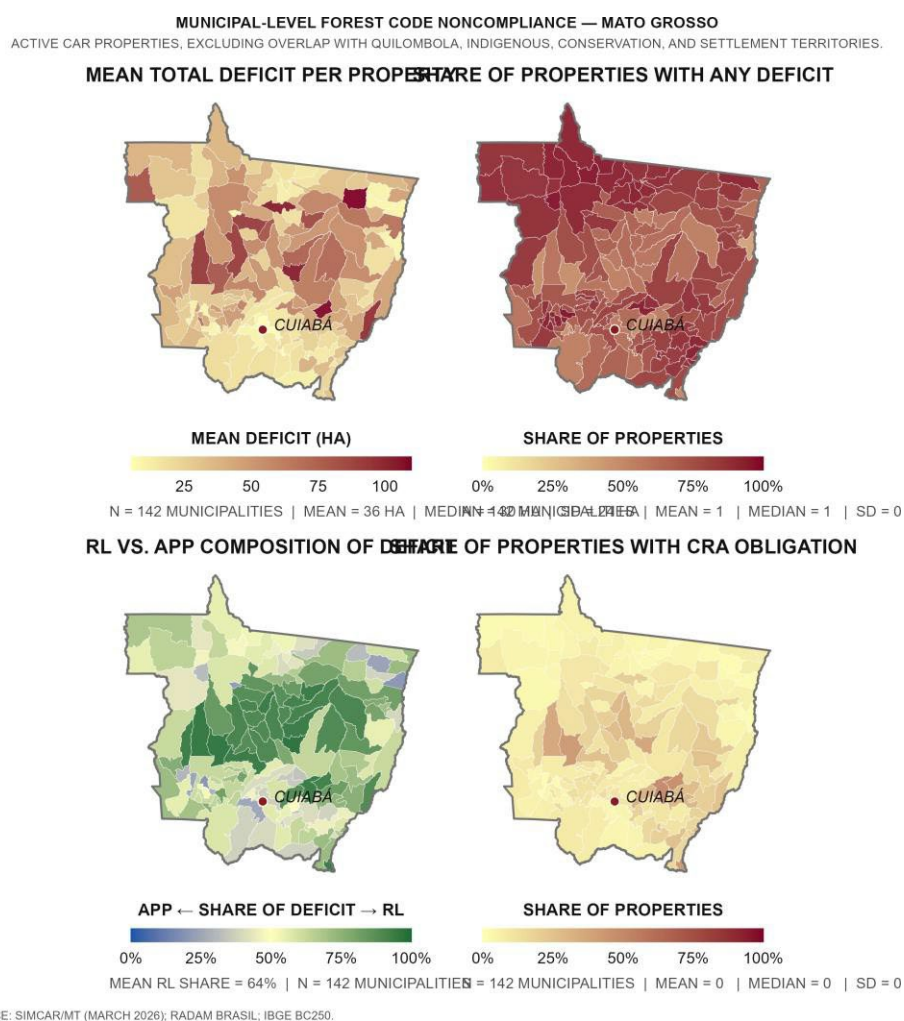


Figure 9: Municipal-level Forest Code noncompliance based on SIMCAR/MT (March 2026), RADAM Basil and IBGE BC250

The spatial analysis of Forest Code compliance—as illustrated in the municipal maps — reveals that the environmental deficit in Mato Grosso has a geographic footprint. Cross-referencing these spatial variables brings a central point: state environmental regularization policies should not be deployed as blanket measures; rather, they require a targeted approach.

A map that shows the share of properties with any deficit demonstrates that noncompliance, in a strictly binary sense, is endemic across nearly the entire state. However, analyzing the first Map (the mean total deficit per property) alters the narrative. It exposes isolated "islands" of liabilities concentrated in the central-north and western corridors of the state. This finding of unequal allocation of forest responsibilities was also found in a previous study, which indicated that approximately 30% of municipalities in Mato Grosso concentrate about 70% of the compensable forest responsibility deficit in the Forest biome (IPAM, 2016).

This distinction is important for optimizing public administration. The bureaucratic heavy lifting required by the state environmental agency (SEMA) to review, validate, and monitor an Environmental Regularization Program (PRA) for a tiny smallholding is virtually identical to the effort needed for a massive corporate estate, especially when using digital systems. Yet, the ecological payoff—measured in actual hectares restored or compensated—varies by orders of magnitude depending on where the intervention takes place.

The data suggests that Forest Code implementation in Mato Grosso is a textbook case for the Pareto principle, which states that, for many outcomes, roughly 80% of consequences come from 20% of causes. If the state were to concentrate its enforcement campaigns, registry validation task forces, and regularization drives on a critical cluster of just 15 to 20 municipalities, the overall statewide deficit would shrink dramatically.

When Map1 is overlaid with Maps 3 (the proportion of the deficit that should be restored in situ) and 4 (the proportion of obligations eligible for CRA compensation), it can be observed that some municipalities exhibit the highest average deficits per property.

Thus, public policy must determine its objective: to concentrate efforts on restoring areas or to activate the compensation market?

If the goal is to allow properties blocked due to market agreements and financial instruments that require compliance with the Forest Code, the policy should focus on areas for restoration. This is because agreements such as the Beef TAC and the Green Grain Protocol of the Federal Prosecutor's Office impose these blocks on properties with deforestation that occurred after 2008. As already stated in another study, this pressure from agreements can serve as a coercive lever to force mass adherence to the PRA in these localities (Gibbs et al., 2016). According to our analysis, 15 municipalities concentrate more than 33% of the areas to be restored, or 823,022 hectares.

Table 16: Municipalities with the largest deficits requiring restoration.

Municipality	Restore (ha)
Colniza	97,141.32
Paranatinga	96,525.23
Juara	80,729.74
Nova Bandeirantes	65,528.21
Nova Maringá	53,879.00
Marcelândia	50,525.72
Nova Uiratã	50,127.19
Aripuanã	44,414.13
União Do Sul	42,930.61
Alta Floresta	42,585.09
Porto Esperidião	40,786.70
Vila Bela Da Santíssima Trindade	40,357.91
Peixoto De Azevedo	39,788.85
Nova Monte Verde	38,871.70
Porto Dos Gaúchos	38,831.03

The spatial distribution of deficits is highly concentrated in a limited number of municipalities. Colniza, Paranatinga, and Juara lead in total area requiring restoration, each exceeding 80,000 hectares of combined LR and APP deficits. This pattern indicates that environmental liabilities are not evenly distributed across the state, but rather clustered in specific geographic areas.

From a policy perspective, this concentration opens opportunities for targeted interventions. Focusing restoration efforts—such as technical assistance, enforcement, and financial incentives—on these municipalities could address a significant share of the state's environmental deficit more efficiently than dispersed actions across the entire territory.

Alternatively, if the option is to boost the number of environmental regularizations through compensation, municipalities with the highest average deficit of LR for compensation may be a good alternative. However, the operation of this market requires properties to register their surpluses and request the issuance of CRA or establish environmental easements, generating effective credit, which can later be used by properties with deficits.

This geographic concentration offers a strategic opportunity for policy implementation, as a coordinated regularization effort focused on these key municipalities and class of properties would resolve a substantial portion of the state's environmental issues. Government and private actions focused on these deficit "hotspots," rather than being dispersed across territory, would optimize human and financial resources, accelerating state compliance metrics.

5.3 Complexity related to secondary vegetation dynamics and the fallow regime

In this chapter, it is essential to address the intrinsic complexities of defining secondary vegetation dynamics. This is because the calculation of compliance for Permanent Preservation Areas and Legal Reserves depends on the amount of existing vegetation on a property—whether primary or secondary—which directly influences the validation of the registry (SEMA, 2024).

Many assessments of FC compliance rely on INPE's PRODES dataset which tracks removal of primary vegetation. However, land use in Brazil is dynamic, with regrowth of secondary vegetation in many cases and sometimes its subsequent removal. Assessing FC compliance considering only primary vegetation runs the risk of resulting in vast overestimates of non-compliance and vast underestimates of natural vegetation cover and possible surpluses. Inclusion of secondary vegetation data was critical for arriving at accurate estimates.

A comparative analysis of geospatial datasets (PRODES data and INPE's Secondary Vegetation data) reveals the existence of 9.24 million hectares of secondary vegetation in Mato Grosso in 2022. This is because, when considering both primary and regenerating vegetation, the total land cover encompasses 59,113,117 hectares. In contrast, excluding secondary forest reduces this value to 49,868,334 hectares (Figure 10).

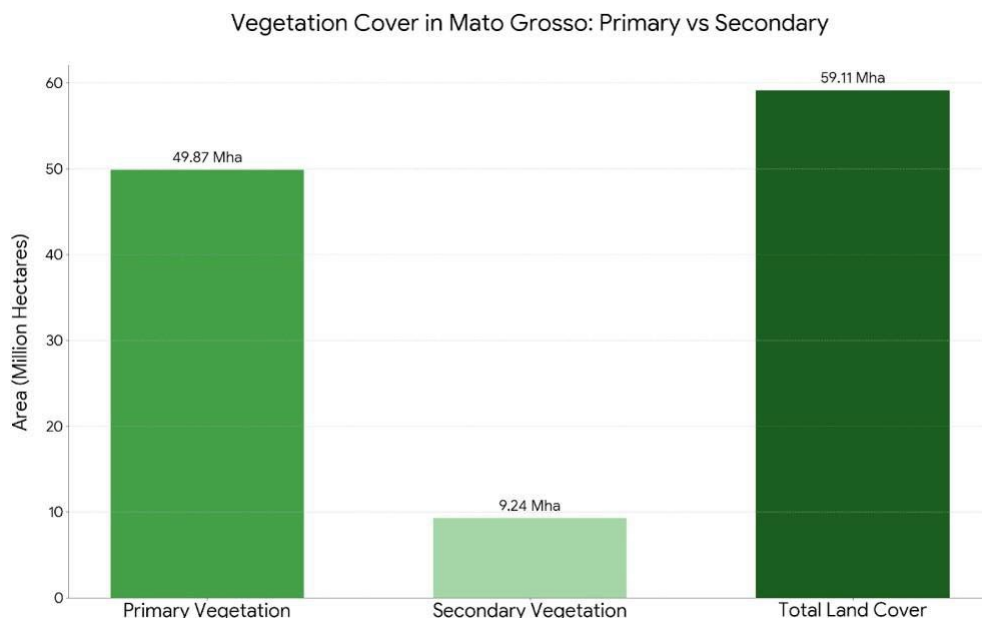


Figure 10: Classification of vegetation cover in Mato Grosso

From a landscape ecology and public policy perspective, while primary forests represent stable carbon stocks and preserved biodiversity, the 9.24 million hectares of secondary forest represent areas of active ecological restoration and land-use transition (Poorter et al. 2021).

There are legal and technical reasons for including secondary vegetation in this study. The Forest Code defines the fallow regime (*regime de pousio*) as a period of up to five years during which land activities are interrupted to allow for the recovery of use capacity or the physical structure of the soil. In Brazilian biomes with denser vegetation, such as the Amazon, Cerrado, and Atlantic Forest, vegetation typically regenerates after this five-year period in areas with natural seed banks (Assunção, 2020). This rapid regrowth, establishing the basis for ecological succession, is classified as secondary vegetation (Assunção, 2020).

Since the vegetation regenerates evidently after this five-year period of disuse, environmental agencies (such as SEMA-MT) legally classify this area as a remnant of native vegetation (in a secondary regeneration stage), making it protected against further unauthorized deforestation (SEMA, 2024).

Therefore, the regulatory framework governing registration and regularization stipulates that

areas that have undergone intervention in the past, but have been abandoned and no longer host productive exploitation, must be legally recognized as areas of remaining native vegetation, encompassing secondary stages of regeneration which, it should be noted, do not lose this protection classification due to eventual fires (CNMP, 2022).

Given this scenario, a major challenge for territorial governance lies in monitoring deforestation in advanced stages of regeneration. Current large-scale monitoring systems do not adequately detect the loss of this secondary vegetation, as official programs such as the Project for Satellite Monitoring of Deforestation in the Legal Amazon (PRODES), by the National Institute for Space Research (INPE), were primarily designed to detect clear-cut deforestation in primary vegetation (Assunção et al., 2020). Once suppression is identified, the area is incorporated into a "historical deforestation mask," making subsequent forest recovery and eventual new deforestation events invisible to official systems (Assunção et al., 2020). This methodological limitation overlooks the fact that secondary vegetation possesses highly complex temporal dynamics, capable of regenerating naturally and being deforested again over time through successive cycles of land abandonment and reoccupation.

The fluid and transitory nature of land cover imposes challenges on the effective implementation of the Forest Code. There is an inherent difficulty in accurately capturing the real status of vegetation within rural properties. Indeed, 15% (more than 9Mha) of the native vegetation detected on Mato Grosso's CAR properties was secondary vegetation, underscoring the importance of including secondary vegetation in monitoring and assessments of FC compliance. The importance of considering temporal dynamics is undeniable, as flora can regenerate and be suppressed again rapidly. In operational practice, this dynamism in land cover can result in cases where validation processes attest to a property's compliance, but the native vegetation declared in the CAR is subsequently lost in the field (CNMP, 2022). Indeed, CAR validations last indefinitely unless they are altered, and SEMA's capacity to monitor changes on properties after validation is limited. Our findings of the substantial role of secondary vegetation highlights the absolute need for continuous and perennial monitoring of rural properties, moving beyond a reliance on "snapshot" analyses to ensure that environmental compliance does not

become an ephemeral condition.

Due to the vast amount of regenerated area in Brazil and increasing state efforts to recover vegetation—driven by recent CAR validation efforts and the implementation of the Environmental Regularization Program (PRA)—it is vital to discuss the inclusion of this refined monitoring in official deforestation alerts. Currently, supply chain compliance protocols, such as the Conduct Adjustment Agreements (TACs) for cattle supply chain promoted by the Federal Prosecutor's Office (often referred to as the "Beef TAC" or Carne Legal), base their audits and commercial restrictions on primary forest deforestation monitoring, excluding offenders from supplier lists through cross-referencing PRODES and land tenure databases (MPF, 2009). The lack of systematic monitoring of secondary vegetation creates a serious loophole where regenerating areas can be successively deforested without triggering commercial blockades for producers. Incorporating secondary deforestation detection into Federal Public Prosecutor's Office protocols and government agency alert systems is fundamental to ensuring the integrity of the Forest Code, guaranteeing that the market and the State protect not only mature forests but also the vast and dynamic ecological capital currently undergoing restoration on rural properties.

5.4 Economic Analysis of Restoration Costs and Environmental Liability Compensation

Once a CAR is validated, the next step in the regularization process is joining the Environmental Regularization Program (PRA). This process imposes significant economic challenges, with costs varying drastically depending on the method adopted to mitigate liabilities in APPs and LRs within rural properties. The analysis of consolidated geospatial data demonstrates an important magnitude of this liability in the territory of Mato Grosso.

For Permanent Preservation Areas, the deficit requiring mandatory restoration on properties reaches 1,792,000 hectares. Regarding Legal Reserves, the net liability totals 4,462,481 hectares. Of this Legal Reserve amount, legislation mandates that 1,020,311 hectares must be restored on-site, as they stem from clearing that occurred after the July 2008 legal milestone, while 3,442,171 hectares are eligible for regularization through off-site compensation mechanisms.

When environmental regularization requires the restoration of vegetation—whether in APPs

or LR—it can occur primarily through two modalities: natural regeneration and active restoration (technically termed recomposition). Natural regeneration consists of a passive ecological recovery process that relies on the ecosystem's own resilience, requiring minimal intervention from the producer (Schimetka et al., 2024). In contrast, active restoration involves the intentional reintroduction of species through seedling planting or direct seeding, demanding the use of machinery, fertilizers and pest control (Brancalion et al., 2019). Which pathway prevails is not arbitrary, but largely determined by the legal status and timing of deforestation. While a portion of liabilities—particularly those associated with recent clearing—must be addressed through restoration within the property, most of Legal Reserve deficits originate from older land-use change and can therefore be resolved through compensation mechanisms.

From a policy perspective, this distinction is critical because the costs associated with these pathways differ by orders of magnitude. Active restoration remains expensive and operationally complex. In the specific context of the Mato Grosso agricultural frontier, recent economic assessments indicate that active forest restoration can require high investments, ranging between \$1,000 and \$3,000 per hectare (Tabaczinski et al., 2025).

On the other hand, natural regeneration presents itself as a considerably more affordable alternative, with average costs near \$60 per hectare (Brancalion et al., 2019; Benini & Adeodato, 2017). However, the adoption of natural regeneration is ecologically restricted by soil condition and the availability of seed sources in the landscape, which often renders its use unfeasible in areas severely degraded by decades of intensive agriculture (Benini & Adeodato, 2017).

As a result, for many producers—especially those managing large deficits on highly productive land—off-site compensation emerges as the most economically viable option. The main instrument designed by the Forest Code for this purpose is the Environmental Reserve Quote (CRA). The market dynamics introduced by this instrument reduce compliance costs for producers in debt and, simultaneously, assign financial value to standing forests, creating a strong monetary incentive for conservation on private lands (Tabaczinski et al., 2025; Soares-Filho et al., 2014). It has been estimated that choosing off-site compensation over on-site restoration could generate savings of approximately one billion dollars for the state of Mato

Grosso alone, avoiding the withdrawal of productive land from the economic system and utilizing existing vegetation stocks (Stickler et al., 2013).

Our findings suggest that Mato Grosso is particularly well positioned to leverage these market-based mechanisms. The state holds a substantial surplus of native vegetation (13,8 million hectares) relative to compensation demand, indicating that, in theory, a large share of historical liabilities could be resolved without extensive new restoration.

Under the terms of the Forest Code, and as validated by the Brazilian Supreme Court in the judgment of ADI 4902, compensation can be carried out between properties located in the same biome without the need for "ecological identity" (nearly identical ecological characteristics for compensation) (STF, 2024). Thus, properties located in different states, but within the same biome, can participate in the compensation market, facilitating this form of payment for environmental services to properties with a vegetation surplus.

However, the effectiveness of this system depends heavily on institutional capacity. Despite the apparent availability of supply, the CRA market remains only partially operational, limiting its role as a cost-reduction mechanism for compliance. Only in October 2025 did the Ministry of the Environment deliver the first quotas for LR compensation and estimated that, nationally, 25.5 million hectares— a market of approximately US\$2.5 billion annually (MMA, 2025)— had the potential for compensation. Also at the end of 2025, through State Decree No. 1,757/2025, the State of Mato Grosso established the State Policy for Legal Reserve Compensation, providing for the creation of the State Environmental Reserve Quota for compensation. However, the specific regulations to operationalize the mechanism and make it available for use by rural landowners are not yet functioning, as connection to and operation of the federal CRA system is necessary (Mato Grosso, 2025). In this context, alternative instruments managed directly by the individual states—such as environmental easements or land acquisition in protected areas—may serve as interim solutions, but do not fully substitute for a functioning market. This gap between legal design and implementation risks delaying regularization and increasing uncertainty for producers.

However, the greatest economic efficiency lies, unequivocally, in preventing illegal

deforestation. The social and private cost of avoiding vegetation suppression is substantially lower than the cost of recovering it after degradation. The land-tenure and environmental clarity provided by CAR validation eliminates legal uncertainty, allowing rural producers to know exactly which areas can be exploited and which must be preserved, avoiding the generation of new liabilities that would require prohibitive investments in future active restoration (Pacheco et al., 2021). The implementation of enforcement policies by environmental agencies and police, using command-and-control strategies guided by territorial intelligence and CAR data, has proven capable of preventing deforestation at a much lower cost than the amounts necessary for the physical restoration of degraded areas (Arima et al., 2014; Assunção, Gandour, & Rocha, 2015).

Beyond restoration, environmental degradation in APP and LR areas generates other costs for the State apparatus, especially regarding enforcement logistics, which requires the deployment of agents to remote regions in the Amazon and the Cerrado. Sector reports indicate that in the Amazon, continental distances and precarious infrastructure raise transport costs and per diems for technical teams, hindering inspection effectiveness and the State's response capacity (Lopes et al., 2025). Data on the historical dynamics of federal inspections reveal that operational inefficiency can exponentially inflate these expenses. Recent analyses show the unit cost for applying sanctions such as embargoes tripled due to excessive spending on logistics and equipment (Oliveira-Filho et al., 2023). In addition to this is the cost of mobilizing the justice system for investigations and processing civil and criminal claims resulting from illegal practices, as well as the costs to the productive sector arising from sanctions.

Therefore, the most efficient strategy for the state of Mato Grosso consists of combining rigorous preventive inspection to halt new liabilities with the facilitation of compensation mechanisms for historical liabilities, such as the CRA, reserving costly active restoration only for areas where there is no legal or ecological alternative.

5.5 The Impact of Environmental Regularization on the Beef Supply Chain and the Need for Individual Traceability

The environmental regularization of the beef production chain represents one of the greatest challenges and, simultaneously, one of the most promising opportunities for territorial governance in Mato Grosso. It is estimated that effective compliance with environmental standards exerts a predominantly positive structural impact on the agricultural sector (Vaccarezza Sevilla et al., 2025). Although it imposes initial transitional and operational costs on rural producers, regularization acts as a decisive catalyst for the sustainable intensification of pastures. This is fundamental to ensuring continued access to highly demanding global markets and mitigating severe commercial risks of exclusion (Stabile et al., 2023). In short, it is an opportunity to build trust in both national and international markets regarding compliance with strict environmental laws by the state's agricultural producers, while stimulating more productive practices and reducing the pressure for new deforestation.

The agreement requires the blocking of properties with deforestation after the FC milestone. However, analyses showed a deficit to recover of 289,093 hectares of LR, which consequently occurred after July 2008 and are potentially illegal because they exceed the area eligible for conversion. These properties that should be blocked continue producing and selling animals to the supply chain, mostly as indirect suppliers.

Table 17: Legal Reserve obligations and compliance in MT cattle chain.

Category	Size Class	Required (ha)	Gross Deficit	Adj. Deficit	To Restore	To Compensate	Surplus (ha)
Direct Supplier	LARGE (>15)	1,080,834	271,961	175,447	17,666	157,781	233,493
	MEDIUM (4-15)	211,357	34,171	63,083	5,469	57,615	35,923
	MINIFUNDIO (≤1)	4,925	5,961	637	565	72	13,345
	SMALL (1-4)	29,814	17,513	1,630	1,435	194	38,401
<i>Subtotal</i>		<i>1,326,930</i>	<i>329,607</i>	<i>240,796</i>	<i>25,135</i>	<i>215,662</i>	<i>321,163</i>
Indirect - T1	LARGE (>15)	1,168,541	259,964	127,585	32,920	94,666	298,389
	MEDIUM (4-15)	314,703	46,033	61,458	13,325	48,132	64,110
	MINIFUNDIO (≤1)	22,522	20,967	5,196	5,072	124	38,209
	SMALL (1-4)	73,564	33,434	6,294	6,096	198	73,481
<i>Subtotal</i>		<i>1,579,330</i>	<i>360,398</i>	<i>200,533</i>	<i>57,412</i>	<i>143,121</i>	<i>474,189</i>
Indirect - T2+	LARGE (>15)	3,551,375	890,566	314,931	70,993	243,938	1,252,476
	MEDIUM (4-15)	1,115,355	153,242	208,195	54,514	153,681	247,014
	MINIFUNDIO (≤1)	180,443	167,545	44,328	43,321	1,007	220,927
	SMALL (1-4)	366,048	180,120	39,731	37,718	2,014	366,583
<i>Subtotal</i>		<i>5,213,221</i>	<i>1,391,473</i>	<i>607,186</i>	<i>206,546</i>	<i>400,640</i>	<i>2,087,001</i>
Total		8,119,480	2,081,478	1,048,515	289,093	759,422	2,882,352

It is important to explain that our estimates of direct and T1 suppliers are conservative because they are based on the TAC rules and just one snapshot (2021-2024), but many of these other properties (currently called Tier 1) likely participate in the supply chain less frequently and were not captured by this particular application of the windows for direct and Tier 1 suppliers. Despite this limitation, the finding that most environmental liabilities occur in indirect suppliers is corroborated by other studies (Rajão et al., 2020; Skidmore et al., 2021).

The same pattern of deficit-to-property size ratio observed in general CAR records also remains here, with large properties exhibiting the highest LR and APP deficit rates.

More than a mere statistical bottleneck, this concentration of non-compliance behind the scenes of production highlights the need for a change in the governance of the cattle chain. Knowing that only a small percentage of properties (7.4%) operate directly with the industry illustrates the enormous scale of the logistical labyrinth that precedes slaughter. This web of intermediary transactions ends up acting, intentionally or not, as a filter that dilutes environmental responsibility even before the meat reaches the shelf.

This contamination dynamic has already been described previously in the beef production chain in the state of Pará, for example, where a link was identified between indirect suppliers and illegal deforestation (Skidmore et al., 2021). The aforementioned research concluded that the lack of monitoring in the initial phases of breeding and rearing operations creates a gap in deforestation control policies. This allows animals raised in deforested areas to be transferred to farms that comply with regulations before being sold to the meat processing industry, introducing irregular products into the formal market through the practice of "cattle laundering" (Skidmore et al., 2021).

One of the points requiring advancement to achieve comprehensive and effective monitoring of the chain is the document that records animal movement: the Animal Transit Permit (GTA). This is a self-declared document primarily designed by the government for sanitary control and batch vaccination (Nassar and Custódio, 2022). Although GTAs are useful for a general analysis

of chain contamination, it is impossible to separate animals individually, as movement is registered by batches and age groups—its main structural limitation. In subsequent sale movements, it is impossible to effectively know the origin of the animals, as a new batch is recorded for each sale, potentially mixing with animals already on the property or arriving from other rural estates. Thus, analyses of the environmental regularity of indirect suppliers—whether in this study or previous works—is needed to estimate possible contamination by illegal deforestation on the farms where the animals were previously located (Rajão et al., 2020; Skidmore et al., 2021). In the context of increasing demands for environmental regularization, driven by the actions of the Federal Prosecutor's Office through the Carne Legal program's Consent Decrees (TACs), the currently necessary reliance on batch-based GTAs facilitates fraudulent practices such as "cattle triangulation" between properties (Gibbs et al., 2025). This systemic vulnerability compromises the sector's reputation, especially in light of recently approved international legislation, such as the European Union Deforestation Regulation (EUDR), which now requires the exact geolocation of all land parcels where cattle were kept throughout their lives and irrefutable proof of the absence of deforestation (Vaccarezza Sevilla et al., 2025).

Individual animal traceability, combined with information on environmental compliance required by cattle agreements and the CAR status for each rural property where the animal has been, is a technological and methodological path capable of ensuring transparency for the links in the chain, thus reducing cattle triangulation. This would allow ranchers, the meat processing industry, and international buyers to verify the animal's compliance throughout the production cycle (Nassar and Custódio, 2022). Furthermore, with individual identification, it is possible to segregate animals that meet specific rules for markets that require more rigorous policies, such as zero deforestation after a certain date, facilitating governance for all actors in the production chain.

In this landscape, certification initiatives such as the Green Passport Program in Mato Grosso, can have a significant positive impact. This program proposes continuous monitoring of

the animal throughout its life cycle using individual identification and an electronic system that centralizes information regarding environmental compliance by the farms (Mato Grosso, 2025).

5.6 The Role of the Green Passport and Digital CAR as a Solution

The progress of environmental regularization in Brazil has historically been hindered by the slow pace of analysis and validation of self-declared data in the CAR. To overcome this structural bottleneck, the implementation of Digital CAR in the state of Mato Grosso represents a technological and regulatory milestone, as the system automates the cross-referencing of producer-declared perimeters with high-resolution thematic cartographic databases (SEMA, 2024). In its current version, the tool has removed the requirement for prior producer consent to issue a validation report, providing agility in identifying forest assets and liabilities (Mariano et al., 2024). This innovation resulted in a significant increase in the volume of completed analyses in the state, allowing the government to quickly quantify deficits in APPs and RL and direct non-compliant properties toward the Environmental Regularization Program (Mariano et al., 2024; Lopes et al., 2025).

However, full compliance will only be achieved with the advancement of instruments that clearly identify the areas to be protected, coupled with a strong signal from the government, society, and the consumer market demanding adherence to the rules (Pacheco and Rajão, 2021). Thus, overcoming the environmental liability in the beef supply chain requires property compliance data to be integrated into the dynamics of cattle movement. The fragmented structure of the livestock sector—divided into cow-calf, backgrounding, and finishing phases—makes indirect suppliers the main vectors of illegal deforestation contamination in the chain, especially given the existing control over direct suppliers through agreements signed with the Federal Prosecutor's Office (Skidmore et al., 2021). Furthermore, animal movement is declared by ranchers through the GTA, which record the movement of batches. While this works well for sanitary control, it leaves gaps for environmental compliance monitoring (Nassar and Custódio, 2022).

In response to this issue, the Green Passport Program (*Passaporte Verde*), established in Mato Grosso and managed by the Mato Grosso Meat Institute (IMAC), seeks to be a solution by linking individual animal traceability to the environmental compliance of the properties (Mato Grosso, 2025). The central objective of the project is the unique identification of each bovine animal from birth, linking the animal's movement history to the compliance of each farm it has passed through during its life cycle. The compliance checklist, to be defined by the program's Steering Committee, will guide what will be checked, and may include a wide variety of verifications, such as compliance with Federal Prosecutor's agreements, the European Union's Free Deforestation Agreement rules, sanitary compliance, CAR status, among others. This systemic integration allows for the full tracking of both direct and indirect suppliers, enabling the issuance of the Socio-Environmental Compliance Certificate (Mato Grosso, 2025).

Specifically regarding the properties that are part of the beef supply chain, our study found that there are 675,362 hectares of total vegetation liability that must be restored across 47,015 properties. Indirect suppliers account for 96% of this liability (650,227 hectares across 40,068 properties). As previously highlighted, existing agreements in this chain historically required compliance only from direct suppliers, and, as the numbers show, this has been successful due to the low number of environmental liabilities in this category of suppliers (Table 19).

Table 18: Net environmental liabilities in cattle suppliers (Direct vs. Indirect).

Category	To Restore (ha)	APP Restore (ha)	To Compensate (ha)
Direct Supplier 2024	25,135	74,541	215,662
Indirect T1 Suppliers	57,412	74,536	143,121
Indirect Supplier T2+	206,546	237,192	400,640
Total	289,093	386,269	759,422

However, a large portion still needs a solution. In this sense, within this operational framework, the Green Passport can act as a catalyst for promoting environmental regularization through its sub-program, the Reinsertion and Monitoring Program (PREM). Properties and rural holdings showing areas illegally deforested after July 2008 are required to join this mechanism

as a condition for being reintegrated into the formal meat market (Mato Grosso, 2025). The process requires the producer to make a formal commitment to isolate and fence off degraded polygons, pay an indemnity fine, and conduct the regeneration of native vegetation on-site through self-inspections and satellite monitoring (IMAC, 2025). In this way, the environmental regeneration process begins immediately, without the need for prior approval of an Environmental Recovery Project (PRA) by environmental agencies, benefiting both the environment and the producer. The law that created the Green Passport also mandates priority for manual CAR analysis by the state environmental agency when necessary.

Another benefit of PREM is the resolution of the producer's legal hurdles to return to a status of legality. Brazilian law provides for administrative and criminal sanctions for illegal deforestation (Chiavari et al., 2023). In cases involving such administrative fines or civil and criminal actions, a type of conciliation table can be established to resolve all sanctions. Here, the party responsible for the property has their CAR analyzed and commits to recovering deforested areas in RL and APPs, reducing the applied fine and fulfilling other obligations adjusted in the agreement, per the interpretation of Article 28-A of the Penal Code, Article 3, Paragraph 2 of the Code of Civil Procedure, and Article 127 of State Supplementary Law 38/1995. Technical cooperation agreements between environmental agencies, the Prosecutor's Office, the State Court, and IMAC can utilize the Green Passport, alongside the Reinsertion and Monitoring Program, as mechanisms to help identify areas eligible for agreements and monitor future environmental recovery, serving as a tool for the swift and effective resolution of pending legal issues.

Although it represents a considerable institutional advance for traceability and environmental regularization in Mato Grosso's livestock sector, the implementation of the Green Passport Program faces structural and socioeconomic bottlenecks, particularly regarding the inclusion of the base of the production chain. The legal requirement to isolate and regenerate illegally deforested areas imposes high transaction and implementation costs in the field (IMAC, 2025). Scientific literature indicates that ecological restoration demands substantial investment and specific technical knowledge for soil preparation, planting, pest control, and managing

competing species (Brancalion et al., 2019). Such resources are often inaccessible to family farmers and small-scale ranchers, who historically face difficulties accessing machinery, credit lines, and adequate technical guidance (Vieira, 2017).

While the legislation creating the Green Passport provides for a government support program—donating inputs and individual identification elements for small properties limited to 100 head of cattle (Mato Grosso, 2025)—the effectiveness of this measure depends on overcoming deficiencies in public technical assistance and rural extension services. Without continuous technical monitoring subsidized by the state or supported by companies and NGOs, there is a risk that these small producers will be unable to meet the technological demands of georeferencing, app-based self-inspections, and individual herd traceability (IMAC, 2025; Imaflora, 2025). Consequently, these vulnerable actors could end up marginalized and excluded from the formal supply chain, worsening socioeconomic inequality and rural poverty in Mato Grosso.

Additionally, the incentive for voluntary entry and retention in the program is hindered by a common cultural and economic perception among rural producers that environmental compliance and the isolation of areas represent an immediate loss of productive territory and business viability (Imaflora, 2025). For the Green Passport to reach the necessary scale of engagement, structuring financial support mechanisms that compensate for the opportunity cost of land is a necessary task. Such mechanisms include Payments for Environmental Services (PES), access to rural credit lines with subsidized interest rates, higher payments for individually identified cattle, and favorable agricultural insurance tied to the producer's environmental performance (Franchini, Viola, and Guivant, 2023; Climate Policy Initiative, 2025). The absence of tangible remuneration for "standing forests" or fair, well-distributed market premiums across the production chain discourages adherence and restricts the reach of regularization primarily to producers with higher capitalization, leaving it dependent on the rigidity of state enforcement and sectoral agreements like TACs.

In the institutional and political sphere, the implementation of rigorous individual traceability and commercial blocking measures faces strong pressure from conservative sectors of

agribusiness (Franchini, Viola, and Guivant, 2023). Historically, the so-called *bancada ruralista* (the ruralist caucus) exerts considerable legislative influence to flex protection norms, delay sanctions, and promote amnesties—as seen during the intense resistance to the 2012 Forest Code and continuous attempts to weaken government control agencies (Rajão et al., 2020). This significant political force often acts to subvert the mandatory nature of socio-environmental compliance, creating a sense of impunity and an expectation of future legal loosening, which serves as a strong deterrent for offenders to assume the immediate costs of entering regularization programs.

Finally, these systemic challenges are compounded by the limitations of the government databases underlying private certification initiatives, coupled with persistent land tenure insecurity characterized by land overlaps, land grabbing (*grilagem*), and a lack of definitive titling (Climate Policy Initiative, 2025; BNL Report, 2025). Without resolving land tenure chaos and accelerating CAR analyses, the guarantee sought by the meatpacking industry and importing markets—that the production chain is indeed free of illegal deforestation—remains structurally weakened.

Despite the significant socioeconomic and structural obstacles, the synergistic action between technological innovations and new institutional arrangements presents a promising horizon. The advent of Digital CAR in Mato Grosso helps break the historic bottleneck of slow validation (Mariano et al., 2024). Complementarily, the Green Passport Program emerges as an engine for transformation by linking individual herd traceability to farm compliance status, providing state-supplied identification tags and assistance for small ranchers to mitigate their technical and financial vulnerability (Mato Grosso, 2023). When paired with the Reinsertion and Monitoring Program, which enables the immediate return of non-compliant properties to the formal market through auditable commitments to isolation and regeneration, the system cools political pressure by proving that legal compliance does not mean the end of profitability. On the contrary, it can even be coupled with projects that increase productivity and farm diversification (IMAC, 2025). When connected to instruments that value standing forests—such as the trading of CRAs and attracting climate-linked investments like carbon credits—these systemic initiatives convert the

burden of regularization into a certificate of sustainability, structuring a transparent, legally secure business environment prepared for the demands of international markets (Vaccarezza Sevilla et al., 2025).

5.7 Consensual and Technological Instruments for Resolving Legal Disputes and Boosting Environmental Regularization

Brazilian legislation allows for the simultaneous and independent application of administrative, civil, and criminal sanctions for illegal deforestation (Chiavari et al., 2023). These procedures are conducted by different state agencies without formal coordination. As a result, landholders may face: (i) administrative embargoes and fines processed by state or federal environmental agencies, depending on which authority issued the citation (Article 72 of Law 9.605/1998); (ii) a civil public action seeking recovery for environmental damages and the application of civil fines (Law 7.347/1985); and (iii) a criminal prosecution for environmental offenses, particularly under Articles 50 and 50-A of Law 9.605/1998.

Our results showed that 43,571 properties have a deficit in LR (Land Use Area) and 128,524 properties have a deficit in APP (Permanent Preservation Area). Due to the triple liability in Brazil, many of these owners may face criminal charges, embargoes, and administrative fines.

The most recent and consolidated official data on the number of civil lawsuits for illegal deforestation were published at the end of 2025 by the JusAmazonia Platform, which extracts and analyzes information directly from the official databases of the National Council of Justice of Brazil (Datajud and SireneJud). Filtering the state of Mato Grosso, the platform indicates that 5,390 cases are being processed in the State Courts and another 692 cases in the Federal Courts, totaling 6,082 Public Civil Actions in progress for this type of environmental damage (JusAmazonia, 2025).

A report produced by the National Council of Justice (CNJ) indicates that the judicial system has been slow in applying laws that promote conservation. The State Court of Mato Grosso has the highest average processing time for socio-environmental actions among all courts in the Legal Amazon; an environmental lawsuit there takes, on average, 2,135 days to be concluded (approximately 7 years). In Federal Court, the average processing time is 1,385 days (approximately

3 years and 9 months) (National Council of Justice, 2022). Judges operating in the Amazon region attribute this procedural lethargy to the high complexity of the cases, overcrowded dockets, and the lack of adequate personnel and budgetary structures for the continuous control and monitoring of judicial compliance (National Council of Justice, 2022).

Additionally, the execution phase of these lawsuits faces factual obstacles, notably the difficulty in locating the party responsible for the violation, the alleged lack of assets to cover fines and restoration costs, and legal maneuvers involving enforcement fraud and the concealment of assets (National Council of Justice, 2022). The long duration of these disputes compromises the State's ability to punish and demand the effective recovery of cleared forests, creating an environment conducive to a sense of impunity and perpetuating degradation in the field.

As a way to resolve this overlap of demands and chronic judicial delay, the Brazilian legal system has begun to encourage the adoption of consensual instruments that facilitate the resolution of disputes and the producer's rapid return to legality. This paradigm shift is supported by Article 3, Paragraph 2, of the Code of Civil Procedure, which dictates that the use of conciliation and mediation should be encouraged by all legal practitioners to resolve conflicts (Brazil, 2015). In the criminal sphere, progress was made through Article 28-A of the Code of Criminal Procedure, which established the Non-Prosecution Agreement (ANPP). This allows the offender to repair the environmental damage and fulfill adjusted conditions in exchange for the non-filing or dismissal of the criminal action (Brazil, 2019). In alignment with these national guidelines, the state of Mato Grosso established amicable settlement mechanisms, notably Article 127 of State Supplementary Law 38/1995 and State Decree 1.436/2022. These regulations instituted the Environmental Fine Conversion Program and created the Environmental Conciliation Nucleus, designing a legal environment conducive to negotiation (Mato Grosso, 1995; Mato Grosso, 2022).

Utilizing this legal framework, "conciliation tables" are formed—often operated during environmental task forces (*mutirões*)—bringing together executive environmental agencies, the Prosecutor's Office, and the Court of Justice in a single setting (SEMA, 2024). In these unified instances, the party responsible for the rural property has the opportunity to have their CAR analyzed swiftly, signing a global agreement to resolve all administrative, civil, and criminal sanctions

weighing on the property at once. By formally committing to recover the illegally deforested area—including the restoration of LR and APPs—the producer gains the benefit of a reduction in the applied administrative fine, ranging from 40% to 90% of the consolidated value, in addition to fulfilling other mitigating and compensatory obligations set in the agreement (Mato Grosso, 2022).

In this context, the Green Passport Program, allied with the Reinsertion and Monitoring Program led by IMAC, can be strategically used by public institutions to expand these conciliation efforts. Technical cooperation agreements between the State Secretariat of the Environment (SEMA), the Prosecutor's Office, the Judiciary, and IMAC can offer integration with territorial intelligence and certification platforms. The Reinsertion and Monitoring Program acts as an auxiliary technological tool for identifying areas eligible for settlement, providing the necessary methodology for isolating degraded polygons and ensuring the continuous future monitoring of environmental recovery through satellite imagery and systemic self-inspections reported by the producer themselves (IMAC, 2025). This resolves one of the major obstacles reported in the execution phase of lawsuits: the tracking of agreement compliance.

This resolution dynamic is becoming essential in the livestock chain because, with the enactment of the Green Passport Law, ranchers with illegal deforestation will have a period of 48 months to implement individual traceability of their animals and enter the Reinsertion and Monitoring Program (Mato Grosso, 2025). It is likely that many of these ranchers are also defendants in civil and criminal actions, in addition to having administrative embargo processes pending at environmental agencies.

The synergy between the State's conciliation tables and the technological apparatus of the Green Passport thus results in a fast and effective resolution instrument. This integrated model allows the backlog of judicial and administrative cases to be resolved through conciliation, while also guaranteeing effective ecological recovery of environmental damage in the field and the safe, transparent, and sustainable reinsertion of the rural producer into livestock supply chains.

5.8 The Global Agenda and Financing the Transition: The Role of Regenerative Landscapes

In the search for solutions to reconcile production and legal compliance, the COP28 Dubai Action Agenda on Regenerative Landscapes emerges as a strategic global framework with direct local

relevance through the Landscape Accelerator Brazil (WBCSD, 2025). This initiative brings together major agribusiness companies, financial institutions, and government actors, and sets an ambitious goal of transitioning large-scale agricultural landscapes towards regenerative practices by 2030, with a priority focus on the Cerrado and Forest biomes—regions central to this research (WBCSD, 2025). The approach is intrinsically linked with the land and environmental regularization challenges identified in Mato Grosso, particularly through its emphasis on recovery of degraded lands as a land-sparing strategy. Estimates suggest that recovering up to 40 million hectares of low-productivity pastures in Brazil could convert these areas into high-productivity agricultural lands, thereby reducing pressure for new deforestation for production expansion (WBCSD, 2025).

This mechanism aligns directly with Brazilian public policies, such as the ABC+ Plan and the Green Path program, creating a synergy that can economically facilitate the restoration of liabilities (WBCSD, 2025). By intensifying production in degraded areas, territory is freed within properties for the native vegetation restoration required by the Forest Code, transforming the environmental liability into a component of a low-carbon, regenerative production system. Additionally, the COP28 initiative addresses one of the most critical points for environmental regularization previously discussed: the cost of restoration. The initiative proposes the use of blended finance to reduce investment risks and mobilize large-scale capital—with the ambition of unlocking US\$ 5 billion by 2030 (WBCSD, 2025). This financial flow is essential to support producers who, although willing to join programs like the Green Passport or Regularization Programs, lack the initial capital for fencing and planting costs.

Although the physical focus on pastures is intensification and the transition to low-carbon agriculture, the funding mobilized by LAB encompasses rural property as an integrated production system, which intrinsically depends on legal compliance to access global markets and attract sustained capital. The initiative's strategic document expressly establishes that one of its priority areas for joint action is strengthening the enforcement of land use laws, with absolute emphasis on compliance with the Brazilian Forest Code, coupled with the need to modernize the CAR to unlock private sector investments (WBCSD, 2025). This corroborates the thesis that CAR validation—whether via streamlined analysis or initiatives like Digital CAR in Mato

Grosso—is not merely a bureaucratic procedure, but a fundamental prerequisite for connecting Brazilian agricultural production to global climate finance flows and markets demanding deforestation-free products.

6. Conclusion

Achieving environmental compliance in Mato Grosso's agricultural supply chains is within reach, but it requires targeted interventions and institutions that support rural producers in resolving their environmental liabilities. This study demonstrates that only 20.4% of active rural properties are in full compliance with the Legal Reserve and Permanent Preservation Area requirements of the Forest Code. The state currently has a net deficit of 4.51 million hectares for Legal Reserve and 1.13 million hectares for Permanent Preservation Area. Of this deficit, 3,18 million hectares can be compensated for off-property and 2.46 million hectares need restoration, which will require an average replanting of 3.9 hectares for minifundio and 235.5 hectares for large properties. The spatial concentration of these liabilities in a small fraction of large properties, combined with a surplus of 13.8 million hectares available for off-site compensation, offers a highly viable and economically efficient path to large-scale regularization. Though most deficits are small, only 8% of properties have initiated the process of resolving environmental liabilities (Environmental Regularization Program).

Despite the progress made in the Federal Prosecutor's Office's cattle agreements, analysis on this production chain reveals that properties with illegal deforestation after 2008 still trade animals, making it necessary to expand monitoring to indirect slaughterhouse suppliers to prevent environmental leakage. Furthermore, illegal deforestation in secondary vegetation remains invisible to monitoring.

To overcome these structural barriers and optimize territorial governance, we suggest the following actions:

- **Accelerate registry validation:** Streamlining and scaling the validation of the Rural Environmental Registry through automated systems, such as the digital registry, are

essential steps for unlocking properties in supply chains and resolving land insecurity.

- **Prioritize targeted public policies:** Environmental regularization efforts should focus on 15 to 20 priority municipalities that hold the largest share of the state's restoration deficit, optimizing human and financial resources.
- **Leverage economic and consensual instruments:** Integrating regulatory enforcement with economic incentives—such as the Green Passport Program, the environmental compensation market, and the consensual resolution of legal disputes—can shift environmental regularization from a compliance burden to a source of economic and reputational value.

Taken together, these complementary technological and institutional strategies offer a viable pathway for reconciling agricultural production with environmental conservation. By adopting such an integrated approach, Mato Grosso can strengthen compliance with the Brazilian Forest Code while consolidating its dual role as a global agricultural leader and a steward of critical forest ecosystems.

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A. Supplemental Data Tables

A.1 Part I: All Properties

Table 19: Number of registered rural properties and total area (ha) by size class.

Category	Size Class	N	% N	Area (ha)	% Area
All Properties	LARGE (>15 MF)	14,827	8.4%	54,648,349	66.4%
	MEDIUM (4-15 MF)	22,084	12.6%	15,387,756	18.7%
	MINIFUNDIO (<=1 MF)	90,928	51.7%	3,618,850	4.4%
	SMALL (1-4 MF)	48,181	27.4%	8,668,590	10.5%
Total	Total	176,020	100.0%	82,323,544	100.0%

Table 20: Legal Reserve (LR) obligations, deficits, and compliance by size class.

Category	Size Class	LR Req	% Req	Grs Def	% Grs	Adj Def	% Adj	Rest.	% Rest	Comp.	% Comp	Surplus	% Sur
All Properties	LARGE (>15 MF)	29,466,758	72.7%	7,371,251	73.9%	2,653,051	58.8%	639,435	48.0%	2,013,616	63.3%	9,224,772	66.7%
	MEDIUM (4-15 MF)	8,005,995	19.8%	1,093,821	11.0%	1,492,575	33.1%	338,409	25.4%	1,154,166	36.3%	1,959,121	14.2%
	MINIFUNDIO (<=1 MF)	742,311	1.8%	571,752	5.7%	157,574	3.5%	152,645	11.4%	4,929	0.1%	855,033	6.2%
	SMALL (1-4 MF)	2,294,553	5.7%	941,245	9.4%	211,858	4.7%	202,966	15.2%	8,892	0.3%	1,793,241	13.0%
Total	Total	40,509,617	100.0%	9,978,070	100.0%	4,515,058	100.0%	1,333,455	100.0%	3,181,603	100.0%	13,832,168	100.0%

Note: LR: Legal Reserve; Grs: Gross; Def: Deficit; Adj: Adjusted; Rest: Restoration; Comp: Compensation; Sur: Surplus.

Table 21: Permanent Preservation Area (APP) obligations, preservation, and deficits by size class.

Category	Size Class	APP Req	% Req	Grs Def	% Grs	Preserv.	% Pres	Rest.	% Rest
All Properties	LARGE (>15 MF)	2,883,806	63.3%	838,876	50.1%	2,044,930	71.0%	715,873	63.3%
	MEDIUM (4-15 MF)	858,649	18.9%	316,643	18.9%	542,006	18.8%	233,701	20.7%
	MINIFUNDIO (<=1 MF)	263,399	5.8%	198,130	11.8%	65,269	2.3%	46,654	4.1%
	SMALL (1-4 MF)	547,441	12.0%	319,709	19.1%	227,732	7.9%	133,924	11.8%
Total	Total	4,553,295	100.0%	1,673,357	100.0%	2,879,938	100.0%	1,130,152	100.0%

Table 22: Combined Forest Code liabilities (LR + APP) by size class.

Category	Size Class	Grs Def	% Grs	Adj Def	% Adj	Rest	% Rest	Comp.	% Comp
All Properties	LARGE (>15 MF)	8,210,127	70.5%	3,368,924	59.7%	1,355,308	55.0%	2,013,616	63.3%
	MEDIUM (4-15 MF)	1,410,464	12.1%	1,726,276	30.6%	572,110	23.2%	1,154,166	36.3%
	MINIFUNDIO (<=1 MF)	769,882	6.6%	204,228	3.6%	199,298	8.1%	4,929	0.1%
	SMALL (1-4 MF)	1,260,954	10.8%	345,782	6.1%	336,890	13.7%	8,892	0.3%
Total	Total	11,651,427	100.0%	5,645,210	100.0%	2,463,606	100.0%	3,181,603	100.0%

A.2 Part II: Special Areas

Table 23: Number of registered rural properties and total area (ha) in Special Areas.

Category	Size Class	N	% N	Area (ha)	% Area
Conservation Units	LARGE (>15 MF)	563	1.1%	2,591,093	40.1%
	MEDIUM (4-15 MF)	825	1.6%	537,573	8.3%
	MINIFUNDIO (<=1 MF)	1,742	3.5%	39,636	0.6%
	SMALL (1-4 MF)	1,259	2.5%	210,260	3.2%
	<i>Subtotal</i>	<i>4,389</i>	<i>8.8%</i>	<i>3,378,563</i>	<i>52.2%</i>
Quilombola Terr.	MINIFUNDIO (<=1 MF)	2	0.0%	2	0.0%
	<i>Subtotal</i>	<i>2</i>	<i>0.0%</i>	<i>2</i>	<i>0.0%</i>
Rural Settlements	LARGE (>15 MF)	72	0.1%	158,722	2.4%
	MEDIUM (4-15 MF)	338	0.7%	220,024	3.4%
	MINIFUNDIO (<=1 MF)	36,821	73.4%	1,629,123	25.2%
	SMALL (1-4 MF)	8,534	17.0%	1,081,178	16.7%
	<i>Subtotal</i>	<i>45,765</i>	<i>91.2%</i>	<i>3,089,047</i>	<i>47.8%</i>
Total	Total	50,156	100.0%	6,467,612	100.0%

Table 24: Legal Reserve (LR) obligations, deficits, and compliance in Special Areas.

Category	Size Class	LR Req	% Req	Grs Def	% Grs	Adj Def	% Adj	Rest.	% Rest	Comp.	% Comp	Surplus	% Sur
Conservation Units	LARGE (>15 MF)	1,323,182	51.9%	543,836	51.8%	16,363	8.7%	6,420	4.4%	9,943	24.6%	859,013	48.7%
	MEDIUM (4-15 MF)	270,972	10.6%	12,457	1.2%	14,747	7.9%	8,748	6.0%	5,999	14.8%	151,630	8.6%
	MINIFUNDIO (<=1 MF)	12,899	0.5%	1,955	0.2%	803	0.4%	698	0.5%	105	0.3%	12,566	0.7%
	SMALL (1-4 MF)	89,152	3.5%	7,079	0.7%	2,578	1.4%	2,471	1.7%	107	0.3%	63,861	3.6%
	<i>Subtotal</i>	<i>1,696,206</i>	<i>66.5%</i>	<i>565,327</i>	<i>53.8%</i>	<i>34,491</i>	<i>18.4%</i>	<i>18,337</i>	<i>12.5%</i>	<i>16,154</i>	<i>39.9%</i>	<i>1,087,069</i>	<i>61.6%</i>
Quilombola	MINIFUNDIO (<=1 MF)	1	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	1	0.0%
	<i>Subtotal</i>	<i>1</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>	<i>1</i>	<i>0.0%</i>
Rural Settl.	LARGE (>15 MF)	86,631	3.4%	9,839	0.9%	12,201	6.5%	1,573	1.1%	10,627	26.2%	24,887	1.4%
	MEDIUM (4-15 MF)	133,184	5.2%	16,070	1.5%	16,948	9.0%	7,104	4.8%	9,844	24.3%	32,357	1.8%
	MINIFUNDIO (<=1 MF)	365,880	14.3%	330,504	31.4%	85,897	45.8%	83,660	56.9%	2,237	5.5%	363,394	20.6%
	SMALL (1-4 MF)	268,529	10.5%	129,081	12.3%	38,101	20.3%	36,456	24.8%	1,646	4.1%	257,255	14.6%
	<i>Subtotal</i>	<i>854,225</i>	<i>33.5%</i>	<i>485,495</i>	<i>46.2%</i>	<i>153,147</i>	<i>81.6%</i>	<i>128,794</i>	<i>87.5%</i>	<i>24,354</i>	<i>60.1%</i>	<i>677,893</i>	<i>38.4%</i>
Total	Total	2,550,431	100.0%	1,050,822	100.0%	187,639	100.0%	147,131	100.0%	40,508	100.0%	1,764,963	100.0%

Table 25: Permanent Preservation Area (APP) obligations, preservation, and deficits in Special Areas.

Category	Size Class	APP Req	% Req	Grs Def	% Grs	Preserv.	% Pres	Rest.	% Rest
Conservation Units	LARGE (>15 MF)	180,173	40.9%	26,210	15.3%	153,962	57.2%	16,832	24.6%
	MEDIUM (4-15 MF)	42,776	9.7%	8,289	4.8%	34,487	12.8%	5,556	8.1%
	MINIFUNDIO (<=1 MF)	3,498	0.8%	1,133	0.7%	2,365	0.9%	214	0.3%
	SMALL (1-4 MF)	16,189	3.7%	3,956	2.3%	12,232	4.5%	1,611	2.4%
	<i>Subtotal</i>	<i>242,635</i>	<i>55.0%</i>	<i>39,588</i>	<i>23.1%</i>	<i>203,047</i>	<i>75.4%</i>	<i>24,214</i>	<i>35.3%</i>
Quilombola	MINIFUNDIO (<=1 MF)	0	0.0%	0	0.0%	0	0.0%	0	0.0%
	<i>Subtotal</i>	<i>0</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>
Rural Settlements	LARGE (>15 MF)	10,103	2.3%	3,775	2.2%	6,328	2.4%	3,496	5.1%
	MEDIUM (4-15 MF)	15,848	3.6%	5,733	3.3%	10,115	3.8%	3,979	5.8%
	MINIFUNDIO (<=1 MF)	104,228	23.6%	77,499	45.1%	26,729	9.9%	20,202	29.5%
	SMALL (1-4 MF)	68,199	15.5%	45,141	26.3%	23,059	8.6%	16,686	24.3%
	<i>Subtotal</i>	<i>198,378</i>	<i>45.0%</i>	<i>132,147</i>	<i>77.0%</i>	<i>66,230</i>	<i>24.6%</i>	<i>44,364</i>	<i>64.7%</i>
Total	Total	441,013	100.0%	171,736	100.0%	269,277	100.0%	68,578	100.0%

Table 26: Combined Forest Code liabilities (LR + APP) in Special Areas.

Category	Size Class	Grs Def	% Grs	Adj Def	% Adj	Rest.	% Rest	Comp.	% Comp
Conservation Units	LARGE (>15 MF)	570,046	46.6%	33,195	13.0%	23,252	10.8%	9,943	24.6%
	MEDIUM (4-15 MF)	20,746	1.7%	20,304	7.9%	14,304	6.6%	5,999	14.8%
	MINIFUNDIO (<=1 MF)	3,087	0.2%	1,017	0.4%	912	0.4%	105	0.3%
	SMALL (1-4 MF)	11,035	0.9%	4,189	1.6%	4,082	1.9%	107	0.3%
	<i>Subtotal</i>	<i>604,915</i>	<i>49.5%</i>	<i>58,705</i>	<i>22.9%</i>	<i>42,551</i>	<i>19.7%</i>	<i>16,154</i>	<i>39.9%</i>
Quilombola	MINIFUNDIO (<=1 MF)	0	0.0%	0	0.0%	0	0.0%	0	0.0%
	<i>Subtotal</i>	<i>0</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>	<i>0</i>	<i>0.0%</i>
Rural Settlements	LARGE (>15 MF)	13,614	1.1%	15,696	6.1%	5,069	2.4%	10,627	26.2%
	MEDIUM (4-15 MF)	21,803	1.8%	20,927	8.2%	11,083	5.1%	9,844	24.3%
	MINIFUNDIO (<=1 MF)	408,004	33.4%	106,100	41.4%	103,863	48.1%	2,237	5.5%
	SMALL (1-4 MF)	174,222	14.2%	54,788	21.4%	53,142	24.6%	1,646	4.1%
	<i>Subtotal</i>	<i>617,643</i>	<i>50.5%</i>	<i>197,511</i>	<i>77.1%</i>	<i>173,157</i>	<i>80.3%</i>	<i>24,354</i>	<i>60.1%</i>
Total	Total	1,222,558	100.0%	256,216	100.0%	215,708	100.0%	40,508	100.0%

A.3 Part III: Active Properties

Table 27: Number of registered rural properties and total area (ha) by size class.

Category	Size Class	N	% N	Area (ha)	% Area
Active Properties	LARGE (>15 MF)	13,156	8.2%	47,988,998	65.9%
	MEDIUM (4-15 MF)	19,949	12.5%	13,836,858	19.0%
	MINIFUNDIO (<=1 MF)	83,491	52.4%	3,294,027	4.5%
	SMALL (1-4 MF)	42,823	26.9%	7,699,686	10.6%
Total	Total	159,419	100.0%	72,819,569	100.0%

Table 28: Legal Reserve (LR) obligations, deficits, and compliance by size class.

Category	Size Class	LR Req	% Req	Grs Def	% Grs	Adj Def	% Adj	Rest.	% Rest	Comp.	% Comp	Surplus	% Sur
Active Properties	LARGE (>15 MF)	25,870,592	72.4%	6,412,403	73.4%	2,328,197	58.2%	521,224	46.4%	1,806,973	62.8%	7,995,210	66.1%
	MEDIUM (4-15 MF)	7,182,688	20.1%	985,048	11.3%	1,355,097	33.9%	295,515	26.3%	1,059,581	36.8%	1,735,296	14.3%
	MINIFUNDIO (<=1 MF)	670,970	1.9%	518,133	5.9%	140,619	3.5%	136,227	12.1%	4,391	0.1%	774,302	6.4%
	SMALL (1-4 MF)	2,015,130	5.6%	819,423	9.4%	179,197	4.5%	171,332	15.2%	7,865	0.3%	1,587,891	13.1%
Total	Total	35,739,380	100.0%	8,735,007	100.0%	4,003,110	100.0%	1,124,299	100.0%	2,878,811	100.0%	12,092,699	100.0%

Table 29: Permanent Preservation Area (APP) obligations, preservation, and deficits by size class.

Category	Size Class	APP Req	% Req	Grs Def	% Grs	Preserv.	% Pres	Rest.	% Rest
Active Properties	LARGE (>15 MF)	2,539,837	62.9%	739,920	49.6%	1,799,917	70.7%	628,187	62.8%
	MEDIUM (4-15 MF)	770,804	19.1%	285,665	19.2%	485,139	19.1%	211,064	21.1%
	MINIFUNDIO (<=1 MF)	239,834	5.9%	180,679	12.1%	59,155	2.3%	42,383	4.2%
	SMALL (1-4 MF)	487,037	12.1%	284,824	19.1%	202,213	7.9%	119,158	11.9%
Total	Total	4,037,511	100.0%	1,491,088	100.0%	2,546,423	100.0%	1,000,792	100.0%

Table 30: Combined Forest Code liabilities (LR + APP) by size class.

Category	Size Class	Grs Def	% Grs	Adj Def	% Adj	Rest.	% Rest	Comp.	% Comp
Active Properties	LARGE (>15 MF)	7,152,323	69.9%	2,956,384	59.1%	1,149,411	54.1%	1,806,973	62.8%
	MEDIUM (4-15 MF)	1,270,713	12.4%	1,566,161	31.3%	506,579	23.8%	1,059,581	36.8%
	MINIFUNDIO (<=1 MF)	698,812	6.8%	183,002	3.7%	178,610	8.4%	4,391	0.1%
	SMALL (1-4 MF)	1,104,247	10.8%	298,356	6.0%	290,490	13.7%	7,865	0.3%
Total	Total	10,226,095	100.0%	5,003,902	100.0%	2,125,091	100.0%	2,878,811	100.0%

A.4 Part IV: Cattle Suppliers

Table 31: Number of registered rural properties and total area (ha) for Cattle Suppliers.

Category	Size Class	N	% N	Area (ha)	% Area
Direct Supplier	LARGE (>15 MF)	598	1.3%	2,257,554	12.6%
	MEDIUM (4-15 MF)	655	1.4%	449,125	2.5%
	MINIFUNDIO (<=1 MF)	1,280	2.7%	53,032	0.3%
	SMALL (1-4 MF)	971	2.1%	180,164	1.0%
	<i>Subtotal</i>	<i>3,504</i>	<i>7.4%</i>	<i>2,939,875</i>	<i>16.4%</i>
Indirect T1	LARGE (>15 MF)	610	1.3%	2,219,370	12.4%
	MEDIUM (4-15 MF)	866	1.8%	620,980	3.5%
	MINIFUNDIO (<=1 MF)	3,636	7.7%	150,145	0.8%
	SMALL (1-4 MF)	1,835	3.9%	332,749	1.9%
	<i>Subtotal</i>	<i>6,947</i>	<i>14.8%</i>	<i>3,323,244</i>	<i>18.6%</i>
Indirect T2+	LARGE (>15 MF)	1,950	4.2%	6,924,790	38.7%
	MEDIUM (4-15 MF)	3,283	7.0%	2,192,741	12.2%
	MINIFUNDIO (<=1 MF)	22,066	46.9%	934,260	5.2%
	SMALL (1-4 MF)	9,265	19.7%	1,581,274	8.8%
	<i>Subtotal</i>	<i>36,564</i>	<i>77.8%</i>	<i>11,633,065</i>	<i>65.0%</i>
Total	Total	47,015	100.0%	17,896,183	100.0%

Table 32: Legal Reserve (LR) obligations, deficits, and compliance for Cattle Suppliers.

Category	Size Class	LR Req	% Req	Grs Def	% Grs	Adj Def	% Adj	Rest.	% Rest	Comp.	% Comp	Surplus	% Sur
Direct Supp.	LARGE (>15 MF)	1,080,834	13.3%	271,961	13.1%	175,447	16.7%	17,666	6.1%	157,781	20.8%	233,493	8.1%
	MEDIUM (4-15 MF)	211,357	2.6%	34,171	1.6%	63,083	6.0%	5,469	1.9%	57,615	7.6%	35,923	1.2%
	MINIFUNDIO (<=1 MF)	4,925	0.1%	5,961	0.3%	637	0.1%	565	0.2%	72	0.0%	13,345	0.5%
	SMALL (1-4 MF)	29,814	0.4%	17,513	0.8%	1,630	0.2%	1,435	0.5%	194	0.0%	38,401	1.3%
	<i>Subtotal</i>	<i>1,326,930</i>	<i>16.3%</i>	<i>329,607</i>	<i>15.8%</i>	<i>240,796</i>	<i>23.0%</i>	<i>25,135</i>	<i>8.7%</i>	<i>215,662</i>	<i>28.4%</i>	<i>321,163</i>	<i>11.1%</i>
Ind. T1	LARGE (>15 MF)	1,168,541	14.4%	259,964	12.5%	127,585	12.2%	32,920	11.4%	94,666	12.5%	298,389	10.3%
	MEDIUM (4-15 MF)	314,703	3.9%	46,033	2.2%	61,458	5.9%	13,325	4.6%	48,132	6.3%	64,110	2.2%
	MINIFUNDIO (<=1 MF)	22,522	0.3%	20,967	1.0%	5,196	0.5%	5,072	1.8%	124	0.0%	38,209	1.3%
	SMALL (1-4 MF)	73,564	0.9%	33,434	1.6%	6,294	0.6%	6,096	2.1%	198	0.0%	73,481	2.5%
	<i>Subtotal</i>	<i>1,579,330</i>	<i>19.4%</i>	<i>360,398</i>	<i>17.3%</i>	<i>200,533</i>	<i>19.1%</i>	<i>57,412</i>	<i>19.9%</i>	<i>143,121</i>	<i>18.9%</i>	<i>474,189</i>	<i>16.4%</i>
Ind. T2+	LARGE (>15 MF)	3,551,375	43.7%	890,566	42.8%	314,931	30.0%	70,993	24.6%	243,938	32.1%	1,252,476	43.5%
	MEDIUM (4-15 MF)	1,115,355	13.7%	153,242	7.4%	208,195	19.9%	54,514	18.9%	153,681	20.2%	247,014	8.6%
	MINIFUNDIO (<=1 MF)	180,443	2.2%	167,545	8.1%	44,328	4.2%	43,321	15.0%	1,007	0.1%	220,927	7.7%
	SMALL (1-4 MF)	366,048	4.5%	180,120	8.6%	39,731	3.8%	37,718	13.1%	2,014	0.3%	366,583	12.7%
	<i>Subtotal</i>	<i>5,213,221</i>	<i>64.2%</i>	<i>1,391,473</i>	<i>66.8%</i>	<i>607,186</i>	<i>57.9%</i>	<i>206,546</i>	<i>71.5%</i>	<i>400,640</i>	<i>52.8%</i>	<i>2,087,001</i>	<i>72.4%</i>
Total	Total	8,119,480	100.0%	2,081,478	100.0%	1,048,515	100.0%	289,093	100.0%	759,422	100.0%	2,882,352	100.0%

Table 33: Permanent Preservation Area (APP) obligations, preservation, and deficits for Cattle Suppliers.

Category	Size Class	APP Req	% Req	Grs Def	% Grs	Preserv.	% Pres	Rest.	% Rest
Direct Supp.	LARGE (>15 MF)	132,823	11.4%	64,756	11.3%	68,067	11.5%	57,553	14.9%
	MEDIUM (4-15 MF)	28,445	2.4%	15,986	2.8%	12,459	2.1%	12,139	3.1%
	MINIFUNDIO (<=1 MF)	4,373	0.4%	3,856	0.7%	516	0.1%	710	0.2%
	SMALL (1-4 MF)	13,124	1.1%	9,855	1.7%	3,269	0.6%	4,139	1.1%
	<i>Subtotal</i>	<i>178,764</i>	<i>15.3%</i>	<i>94,453</i>	<i>16.4%</i>	<i>84,311</i>	<i>14.2%</i>	<i>74,541</i>	<i>19.3%</i>
Ind. T1	LARGE (>15 MF)	137,615	11.8%	53,611	9.3%	84,004	14.2%	47,978	12.4%
	MEDIUM (4-15 MF)	44,146	3.8%	22,092	3.8%	22,054	3.7%	16,578	4.3%
	MINIFUNDIO (<=1 MF)	13,068	1.1%	11,099	1.9%	1,969	0.3%	2,399	0.6%
	SMALL (1-4 MF)	26,668	2.3%	18,506	3.2%	8,162	1.4%	7,582	2.0%
	<i>Subtotal</i>	<i>221,496</i>	<i>19.0%</i>	<i>105,308</i>	<i>18.3%</i>	<i>116,188</i>	<i>19.6%</i>	<i>74,536</i>	<i>19.3%</i>
Ind. T2+	LARGE (>15 MF)	404,913	34.7%	146,699	25.5%	258,213	43.6%	129,147	33.4%
	MEDIUM (4-15 MF)	155,591	13.3%	74,591	13.0%	81,000	13.7%	55,587	14.4%
	MINIFUNDIO (<=1 MF)	80,610	6.9%	66,869	11.6%	13,741	2.3%	16,352	4.2%
	SMALL (1-4 MF)	125,648	10.8%	87,236	15.2%	38,412	6.5%	36,106	9.3%
	<i>Subtotal</i>	<i>766,761</i>	<i>65.7%</i>	<i>375,395</i>	<i>65.3%</i>	<i>391,367</i>	<i>66.1%</i>	<i>237,192</i>	<i>61.4%</i>
Total	Total	1,167,021	100.0%	575,155	100.0%	591,866	100.0%	386,269	100.0%

Table 34: Combined Forest Code liabilities (LR + APP) for Cattle Suppliers.

Category	Size Class	Grs Def	% Grs	Adj Def	% Adj	Rest.	% Rest	Comp.	% Comp
Direct Supp.	LARGE (>15 MF)	336,718	12.7%	232,999	16.2%	75,219	11.1%	157,781	20.8%
	MEDIUM (4-15 MF)	50,157	1.9%	75,223	5.2%	17,608	2.6%	57,615	7.6%
	MINIFUNDIO (<=1 MF)	9,818	0.4%	1,347	0.1%	1,275	0.2%	72	0.0%
	SMALL (1-4 MF)	27,368	1.0%	5,768	0.4%	5,574	0.8%	194	0.0%
	<i>Subtotal</i>	<i>424,060</i>	<i>16.0%</i>	<i>315,337</i>	<i>22.0%</i>	<i>99,676</i>	<i>14.8%</i>	<i>215,662</i>	<i>28.4%</i>
Ind. T1	LARGE (>15 MF)	313,575	11.8%	175,563	12.2%	80,897	12.0%	94,666	12.5%
	MEDIUM (4-15 MF)	68,125	2.6%	78,036	5.4%	29,904	4.4%	48,132	6.3%
	MINIFUNDIO (<=1 MF)	32,066	1.2%	7,595	0.5%	7,471	1.1%	124	0.0%
	SMALL (1-4 MF)	51,940	2.0%	13,876	1.0%	13,678	2.0%	198	0.0%
	<i>Subtotal</i>	<i>465,706</i>	<i>17.5%</i>	<i>275,069</i>	<i>19.2%</i>	<i>131,949</i>	<i>19.5%</i>	<i>143,121</i>	<i>18.9%</i>
Ind. T2+	LARGE (>15 MF)	1,037,265	39.0%	444,078	30.9%	200,140	29.6%	243,938	32.1%
	MEDIUM (4-15 MF)	227,833	8.6%	263,782	18.4%	110,100	16.3%	153,681	20.2%
	MINIFUNDIO (<=1 MF)	234,414	8.8%	60,681	4.2%	59,674	8.8%	1,007	0.1%
	SMALL (1-4 MF)	267,356	10.1%	75,837	5.3%	73,824	10.9%	2,014	0.3%
	<i>Subtotal</i>	<i>1,766,867</i>	<i>66.5%</i>	<i>844,378</i>	<i>58.9%</i>	<i>443,738</i>	<i>65.7%</i>	<i>400,640</i>	<i>52.8%</i>
Total	Total	2,656,633	100.0%	1,434,784	100.0%	675,362	100.0%	759,422	100.0%